Supplementary Methods, Results and Reproducibility

Companion to "Predictive Contributions of Local Spike History in Mouse Primary Motor Cortex: A Development and Independent-Animal Descriptive Evaluation."

Scientific status: Development exploratory; independent-animal Final DESCRIPTIVE ONLY. The original positive-claim conjunction was not met. This document reports saved evidence; it is not a new neural analysis, external peer review or a second Final assessment. Main Figures 1-7 and all CSV supplements are part of the package.

S1. Source versions, original cohort and attrition

The provider is the International Brain Laboratory. The actual neural source is native ONE/ALF in the public AWS bucket `ibl-brain-wide-map-public`, region `us-east-1`, under CC BY 4.0. The source's Brainwidemap documentation describes 139 mice, 459 sessions and 699 insertions. The analysis is a small MOp subset; those global numbers are not this study's sample size. Relevant original task and source publications and official licence are cited in the main references [1,2,10-12]. Access dates in the bibliography refer to document verification, not a new data release.

The metadata eligibility screen used `2024_Q2_IBL_et_al_BWM` cluster/trial aggregates and `2023_12_bwm_release` insertions. Native sorting was `alf/{probe}/pykilosort/#2024-05-06#/`; the native trial table was `alf/#2025-03-03#/_ibl_trials.table.*.pqt`. Native wheel files were uniquely matched unrevisioned `alf/_ibl_wheel.*` objects. Original sync, AP headers and `_iblqc_ephysSaturationAP.samples` came from the selected probe's raw-ephys collection. Unrevisioned objects were still individually pinned by UUID, ETag, byte size and source/header identity. No latest-version fallback, other-probe substitution or DANDI conversion was used. Prior data-source packages retain native inventory details, not an invented monolithic new release number.

Provider mapping definitions were audited against frozen ibllib revision `3eced8ed26c2dfb6d149472c7ff2246f27d2bfb3`; the old eligibility loader source was pinned at `118fc36cb3602934466ad2c6087c2b3b441f9f1f`. Original native access used ONE-api3.6.1 public unsigned S3 and ALF readers; a prior anonymous metadata-query403 did not cause a dataset substitution. No raw AP/LFP, waveform or video analysis was performed in the predictive study. Header/support checks are not a re-extraction of the provider's spike sorting.

Original fixed allocation:20 mice,12 Development and8 Final. Ten Development/eight laboratories and six Final/four laboratories were admitted. Development exclusions were NYU-37(B=31<40) and CSHL059(sync domain[0,1]s; zero covered epochs). Final exclusions were NYU-21(non-strict wheel timestamps;16,485 equal adjacent pairs) and UCLA033(6 C-qualified units<8). No replacement mouse/session/probe or eligibility definition was introduced. The following inventory gives every original mouse, lab, date, session, probe, eid and pid; long identifiers are placed on separate lines for readability. The exact CSV is `tables/SOURCE_COHORT_IDENTITIES.csv`.

NYU-37 - DEVELOPMENT; angelakilab; 2021-01-26; session 1; probe01. Admitted: False. Exclusion: 31 B epochs, below40 frozen minimum.

Session eid: `f88d4dd4-ccd7-400e-9035-fa00be3bcfa8`

Probe pid: `decc8d40-cf74-4263-ae9d-a0cc68b47e86`

CSHL059 - DEVELOPMENT; churchlandlab; 2020-03-03; session 1; probe00. Admitted: False. Exclusion: sync domain[0,1] yields zero covered epochs.

Session eid: `d2f5a130-b981-4546-8858-c94ae1da75ff`

Probe pid: `79628a45-c2e3-4206-97df-4c91edaff90f`

UCLA037 - FINAL_RESERVED; churchlandlab_ucla; 2022-02-09; session 1; probe00. Admitted: True. No exclusion.

Session eid: `8d316998-28c3-4265-b029-e2ca82375b2f`

Probe pid: `5245b505-d0e7-4942-94d1-b33a084fab03`

KS046 - DEVELOPMENT; cortexlab; 2020-12-05; session 1; probe00. Admitted: True. No exclusion.

Session eid: `aa3432cd-62bd-40bc-bc1c-a12d53bcbdcf`

Probe pid: `41ea689a-7e05-463b-915a-90e63b8c0572`

DY_014 - FINAL_RESERVED; danlab; 2020-07-17; session 1; probe00. Admitted: True. No exclusion.

Session eid: `bd456d8f-d36e-434a-8051-ff3997253802`

Probe pid: `d258bca2-bf18-4491-8d57-5322f705caec`

PL033 - DEVELOPMENT; hausserlab; 2022-10-04; session 2; probe00. Admitted: True. No exclusion.

Session eid: `62902992-8432-46fb-af12-6392012e58c7`

Probe pid: `edc82c06-83fd-44be-a41e-52fdeea728b4`

SWC_043 - DEVELOPMENT; hoferlab; 2020-09-19; session 1; probe00. Admitted: True. No exclusion.

Session eid: `88d24c31-52e4-49cc-9f32-6adbeb9eba87`

Probe pid: `3fd2d2b3-9415-4b6e-b392-40427fb1c388`

ZFM-01576 - FINAL_RESERVED; mainenlab; 2020-12-03; session 1; probe00. Admitted: True. No exclusion.

Session eid: `9e9c6fc0-4769-4d83-9ea4-b59a1230510e`

Probe pid: `6eb8be8d-d089-43c6-b2e4-a6558ca16dcf`

SWC_038 - DEVELOPMENT; mrsicflogellab; 2020-08-01; session 1; probe01. Admitted: True. No exclusion.

Session eid: `03063955-2523-47bd-ae57-f7489dd40f15`

Probe pid: `7bd5627e-b02e-47f1-b476-4bf8eaa726b3`

NR_0027 - FINAL_RESERVED; steinmetzlab; 2022-08-23; session 1; probe01. Admitted: True. No exclusion.

Session eid: `ae8787b1-4229-4d56-b0c2-566b61a25b77`

Probe pid: `310c60b6-d68f-4018-a86a-3668ce296837`

CSH_ZAD_026 - DEVELOPMENT; zadorlab; 2020-08-17; session 1; probe01. Admitted: True. No exclusion.

Session eid: `81a78eac-9d36-4f90-a73a-7eb3ad7f770b`

Probe pid: `03d2d8d1-a116-4763-8425-4ef7b1c1bd35`

NYU-30 - DEVELOPMENT; angelakilab; 2020-10-20; session 1; probe00. Admitted: True. No exclusion.

Session eid: `77e6dc6a-66ed-433c-b1a2-778c914f523c`

Probe pid: `7b05cccc-44f6-4491-a0ea-e38d6e95513d`

UCLA033 - FINAL_RESERVED; churchlandlab_ucla; 2022-02-17; session 1; probe00. Admitted: False. Exclusion: FEWER_THAN_8_C_QUALIFIED_UNITS.

Session eid: `5455a21c-1be7-4cae-ae8e-8853a8d5f55e`

Probe pid: `e96cbd77-5fdf-4a35-81f4-4a6361bded65`

KS051 - DEVELOPMENT; cortexlab; 2021-05-13; session 1; probe01. Admitted: True. No exclusion.

Session eid: `7082d8ff-255a-47d7-a839-bf093483ec30`

Probe pid: `2122d807-13ab-494e-a1cc-e55cf24d3e9f`

DY_011 - FINAL_RESERVED; danlab; 2020-01-31; session 1; probe00. Admitted: True. No exclusion.

Session eid: `7bee9f09-a238-42cf-b499-f51f765c6ded`

Probe pid: `26118c10-35dd-4ab1-9f0f-b9a89a1da070`

ZFM-01937 - DEVELOPMENT; mainenlab; 2021-04-08; session 1; probe00. Admitted: True. No exclusion.

Session eid: `113c5b6c-940e-4b21-b462-789b4c2be0e5`

Probe pid: `af2a0072-e17e-4368-b80b-1359bf6d4647`

CSH_ZAD_019 - DEVELOPMENT; zadorlab; 2020-08-16; session 1; probe00. Admitted: True. No exclusion.

Session eid: `5adab0b7-dfd0-467d-b09d-43cb7ca5d59c`

Probe pid: `b4131145-9675-4433-88e1-9ae016c701dd`

NYU-21 - FINAL_RESERVED; angelakilab; 2020-08-13; session 2; probe00. Admitted: False. Exclusion: NATIVE_WHEEL_NOT_STRICTLY_INCREASING.

Session eid: `8c33abef-3d3e-4d42-9f27-445e9def08f9`

Probe pid: `f475ae14-9415-453e-b800-1480ea1c868d`

UCLA035 - DEVELOPMENT; churchlandlab_ucla; 2022-02-18; session 1; probe01. Admitted: True. No exclusion.

Session eid: `6f36868f-5cc1-450c-82fa-6b9829ce0cfe`

Probe pid: `7c94d733-b913-4064-83f2-37422712204c`

ZFM-01936 - FINAL_RESERVED; mainenlab; 2021-01-20; session 1; probe00. Admitted: True. No exclusion.

Session eid: `4aa1d525-5c7d-4c50-a147-ec53a9014812`

Probe pid: `fddceb2e-82ba-493d-baba-c5d8af65981f`

S2. Source support, native anatomy and unit QC

Anatomy was native MOp parent/layers1,2/3,4,5,6a,6b only. Map cluster channels to native channel atlas entries; do not positionally match an aggregate region array. Source/native annotation differences are retained in the original metadata records. Source label1 is upstream provider-good curation over the recording, not an A/C-only metric. No layer-specific causal or histological-distance claim was registered.

AP duration, sample rate, saved channel count and source header/file-size accounting defined recording support. Source sync was finite, Nx2 and strictly increasing in both clocks. Piecewise-linear probe-to-session mapping followed the provider convention and did not extrapolate eligible epochs. Authorized-role spike times were finite, nondecreasing and inside support to one AP sample. Essential inconsistencies stopped admission; they were not dropped or converted to zeros. Invalid/saturation intervals were mapped from sample bounds through the same clock. Only known selected-probe annotations were available; absence of another annotation is unknown artifact burden, not a guarantee of lossless or artifact-free acquisition.

Physiology used A or C warm-up-plus-score exposure, with no ISI crossing an epoch/gap. Required rate0.5-100Hz inclusive,>=100spikes,>=99 adjacent within-epoch ISI pairs, refractory fraction(ISI<1ms)<=0.005; each original A/C quartile required>=10s exposure and>=0.1Hz. Rank by refractory fraction ascending, rate descending, UUID lexical. Primary first8 units were fixed before B/E/D scoring. Final source/C totals and all48 selected unit IDs/UUIDs with C-only metrics follow; achieved quartile values are retained exactly in the companion CSV, not inferred from Final D.

Mouse	C epochs	D metadata epochs	C-qualified	Admitted
UCLA037	222	171	169	True
DY_014	335	218	49	True
ZFM-01576	601	402	29	True
NR_0027	484	191	17	True
UCLA033	263	154	6	False
DY_011	227	154	17	True
NYU-21	not read	not read	not read	False
ZFM-01936	336	224	66	True

Mouse/rank	Cluster ID	Selected UUID	C Hz	ISI fraction
UCLA037/1	174	3f1a042f-c9a0-4a4c-a2b8-1c9b15f56ac6	71.68919	0
UCLA037/2	506	8c5f533b-2f67-466f-a578-9206f5138335	49.84685	0
UCLA037/3	68	75ac4123-0520-49f7-ac09-72a02219d441	39.93694	0
UCLA037/4	234	35a1c54a-e339-44bd-a5c9-c42f9b7fbfe4	37.02703	0
UCLA037/5	35	af921fbd-50ab-4ed3-afab-4ee7e45b120d	23.75676	0
UCLA037/6	387	53760695-9471-44a3-a831-76870f9884d9	20.54054	0
UCLA037/7	349	876b6a1a-6fe1-4dd0-b011-1652f6efb783	19.62162	0
UCLA037/8	558	ea8c0b64-364c-4bba-8739-37dec45347e3	19.48198	0
DY_014/1	516	9f801920-7ad3-48d8-8f56-5614fab4b2cd	48.82985	0
DY_014/2	512	70e70cac-416e-41f0-8130-864a56753e59	30.53134	0
DY_014/3	475	8301a68c-1188-4623-93c2-a2eaa9c58b22	20.69552	0
DY_014/4	557	837d26bc-4d00-4664-8c9b-65b62027d901	19.80896	0
DY_014/5	436	0d872743-4a9f-4fa5-a176-df8016ef0d09	16.83881	0
DY_014/6	568	a4c5418d-4f4b-453e-a25d-a8a368df3465	15.60597	0
DY_014/7	511	c8606f83-b535-49f2-b9aa-585d97ea0ea0	15.53731	0
DY_014/8	561	4be80f2b-5114-42c1-8ebc-9ebbde69c0eb	15.15224	0
ZFM-01576/1	361	5294eedb-36a1-4b35-b0da-b3eac35aa23a	17.18636	0
ZFM-01576/2	409	d3976615-c44a-4013-944a-e7ba50475409	14.13977	0
ZFM-01576/3	410	1c091af3-9f35-4dae-929d-a04fcca8ff1f	13.50083	0
ZFM-01576/4	355	4035f8e5-d057-4c70-9511-a13f30e2337b	6.575707	0
ZFM-01576/5	442	251ae6cc-9060-49df-af73-7dbaf4c3fe4e	5.44426	0
ZFM-01576/6	365	694534c9-0692-4846-88c5-2c4626c2a0ea	5.069884	0
ZFM-01576/7	400	3d745e26-768a-4155-9266-46bf2accb861	4.104825	0
ZFM-01576/8	372	02c5be99-098a-4e72-b738-86a36f19ac8d	3.745424	0
NR_0027/1	398	4dff9d81-8f12-441f-ac7a-001271f97d90	7.553719	0
NR_0027/2	469	a02d8b2e-2bbb-4146-be2c-c37609ec5a5d	6.237603	0
NR_0027/3	384	6003e131-a4c6-4100-8a59-54fa641db907	5.183884	0
NR_0027/4	368	13a1235d-7f8d-4d9e-9337-d217f2f937b0	4.020661	0
NR_0027/5	768	a0aef07f-9539-4f4d-a216-b401a01a87ea	1.801653	0
NR_0027/6	478	78f54d3a-7e73-45c7-a003-b1f29869794b	1.692149	0
NR_0027/7	404	8e5e647b-1ca8-4bf3-976c-00cd2b020339	1.692149	0
NR_0027/8	356	47234cc4-2f2e-43c5-adcf-7b7cd455cc3e	1.545455	0
DY_011/1	451	f99bb337-2301-4f09-b364-a88edac0e6d8	36.44493	0
DY_011/2	436	05a19746-fd06-422b-8d36-fe66fdb7f55e	9.242291	0
DY_011/3	434	59ed37ff-4f1f-4658-9aa8-757c101ba6f6	7.973568	0
DY_011/4	383	f68e3d9a-864d-414c-ad4c-44af05e26f73	4.39207	0
DY_011/5	568	3157dcaf-9e70-4e36-b395-e38a7da8fe5b	3.651982	0
DY_011/6	571	aec45486-7a59-4524-93ca-bc22a69a2538	3.599119	0

Mouse/rank	Cluster ID	Selected UUID	C Hz	ISI fraction
DY_011/7	423	2cccc7c6-2f2b-466f-a973-874e7ee9a9c7	1.837004	0
DY_011/8	580	f918decf-0572-42dc-bdda-48065ff54101	1.682819	0
ZFM-01936/1	605	e1750c6b-84fc-49a1-870f-a2643607244d	31.42262	0
ZFM-01936/2	625	812b9053-cae4-4f32-ba9c-18fd7b53cc12	26.71429	0
ZFM-01936/3	930	bd9471e7-0d36-4f1d-9ea1-8b506ab4c015	17.91071	0
ZFM-01936/4	590	42048ed9-f4dd-4bbb-b14e-5415b576f48c	12.38095	0
ZFM-01936/5	589	eb501e08-0ac5-4f80-b7ea-88e36a500a57	12.05357	0
ZFM-01936/6	517	11c62a4e-c880-4ba8-a194-fa718a1a4905	9.931548	0
ZFM-01936/7	708	039e9921-ca1c-4881-a9b9-302b60b4ce19	9.491071	0
ZFM-01936/8	632	e9308ba8-3ff0-43d3-bdf2-ed1c75fbbf25	9.136905	0

The full CSV retains C spike counts, exposure, all four quartile rates/exposures and rank. C-level summaries include UCLA033's source-pass/QC-fail state. NYU-21 had no C-neural projection because its source gate failed. Main Figure7 and `FINAL_ANATOMY_LAYER_PROXY.csv` retain primary/sensitivity anatomy and root/unknown categories; native depth was unavailable. Neighboring contact-index checks use±2 positions and are not physical distance estimates.

S3. Trial roles, observation masks and strictly past support

For original native trial count N, Development cuts are floor(0.6N) and floor(0.8N); Final cut is floor(0.6N). Role intervals are half-open original-row ranges. Drop the last trial before and first trial after each cut from adjoining roles. Never reindex or recut after invalid rows. Native ordinal, event values, missingness, saturation clock and role masks are frozen in `paper15_r1/metadata/*_task.json` and `paper15_final1/metadata/*_task.json`; original source hashes are in SOURCE_HASHES.json. The full original masks remain audit inputs, not regenerated from a guessed trial count.

A primary epoch spans stimulus-0.5s through stimulus+0.5s, with ten50-ms warm-up and ten score bins. Its neural interval lies within the trial/support/sync domain and cannot overlap another epoch. Finite start/stim/goCue/feedback/end obey source ordering(start<=stim<=feedback<=end; goCue inside trial before feedback), without an arbitrary cue/stim ordering assumption. Known invalid or saturation overlap excludes the whole epoch for all models. For behavioral predictors, the extra predecessor support spans stimulus-1s through stimulus+0.5s and lies within the permitted chronological role and recording. Neural history still resets at epochStart; behavioral predecessors may precede the current trial. Zero counts in valid bins remain observations.

All events/samples satisfy time<binStart before contributing. Equality is unavailable at the boundary. Wheel positions are backward sample-and-hold, not interpolated through future observations. Current choice/reward appears only after its source event. Prior completed-trial codes are frozen at epochStart and use already observed task events. A stimulus exactly at a bin start enters only later predictor rows; stimulus-aligned analysis windows are an offline sampling design, not a guarantee of live prediction at unknown future stimuli.

Minimum Development admission was80A/40B/40E epochs and8units, with>=8mice/>=4labs. Final needed80C/60D metadata-valid epochs and8C-qualified units;>=8mice was required for confirmation,6-7 only descriptive,<6 stopped D. D-neural quality/score was not an eligibility screen. Any essential integrity failure discovered during the one batch could not trigger replacement, recalibration or retry. The actual six-mouse fallback was fixed before observing D results.

S4. Native binary64 correction and complete attempt ledger

The original Development source implementation serialized native numeric task data with ten decimal places. That altered strict-past wheel-derived inputs in5/10admitted mice; the execution stopped at its source-implementation gate. Original358 completed fits had numerical endpoint certificates on rounded inputs, one STARTED attempt lacked an endpoint, and251 planned main fits were unattempted. No original coefficient was reused as a corrected fit, and numerical success was not called scientific source validity.

A separately prospective corrective mission retained exact binary64 task values and native absent-side missingness. Source audit checked96,642finite numeric table cells(not distinct timestamps) and382,487strict predecessor queries. PL033 trial224's equal-time wheel event was excluded under strict<. The corrected Development study completed610main+525sensitivity=1,135starts. Historical359+corrective1,135=1,494before Final; Final72+320=392; full historical total1,886. The original aggregate ceiling1,928remains in the frozen protocol; separate corrective and Final authorizations are disclosed rather than claiming retrospective budget compliance. Final explicit caps were96main/440sensitivity; no retry or post-D fitting occurred.

The earlier source failure, numerical stops, review failures, partial execution and subsequent archive/self-log formatting incidents remain in the old evidence directories. Their exact bytes/hashes are preserved. Writing this paper does not retrospectively relabel any failed scientific gate. It introduces no optimizer rescue or new data.

S5. Full predictor definitions and transforms

Let b be binStart; I(condition) its indicator; clip(a, L)=max(0, min(a, L)). Unobserved events have zero indicator/elapsed/value columns. The following are the37 non-intercept nuisance columns before training-only standardization. Values are source observations available strictly before b.

Columns	Raw definition
1-2	e=clip(b-trialStart,10)/10 and e squared
3-5	r=I(stim<b)clip(b-stim,1), r squared, r cubed
6-7	I(stim<b) times left/right contrast; source-absent side becomes0
8-9	I(stim<b) times left/right positive-contrast indicator
10-11	Columns6/7 multiplied by r
12-13	I(goCue<b), I(goCue<b)clip(b-goCue,1)
14-17	I(response<b), signed choice times response indicator, I(feedback<b), feedback type times feedback indicator
18-19	Reward code and signed choice of most recent completed trial with corresponding event before epochStart; frozen at reset

Columns	Raw definition
20-21	Original trial ordinal/1000 and its square
22-23	Session time b/3600 and its square
24	Backward-held wheel position at b minus position at epochStart, radians
25-28	Signed past displacement over0.05,0.1,0.25,0.5s
29-32	Total absolute observed encoder increments over the same four intervals, including the preceding held value
33-36	Squared signed wheel velocity: (displacement/h)^2, for each interval
37	Indicator absolute0.05-s displacement>0.001radians

For columns33-36, the exact operation is (signed displacement/h) squared. No current-bin response or retrospective firstMovement estimate is used. Source-NaN on exactly one contrast side means absent and is mapped to0; both absent/unknown or a finite value outside[0,1] excludes that trial. Trial ordinal is not normalized by the future trial count. Source previous-trial codes are behavior, not inherited neural state.

Four disjoint neural windows are [b-0.05, b),[b-0.1, b-0.05),[b-0.2, b-0.1),[b-0.5, b-0.2). Count/windowDuration gives Hz, with all outside-epoch neural history zero. Four features per unit create4U history columns. No within-bin target count is a predictor.

Nonconstant ordinary inputs use A/C score-row population means and SDs, then±3clipping. SD<1e-8 gives an identically zero input and a fixed-zero coefficient; intercepts are not standardized. Persist nominal/active maps, scales, units, trial IDs and masks. C first standardizes microbin rates by A/C score-row mean/SD without clipping; its projected history q then receives a separate A/C score-row transform and±3clipping. These are feature transforms, not predicted-rate or generated-count clipping.

S6. Full model equations, gauges and parameter counts

For joint population bins, Poisson exposure is explicit and factorials remain in reported likelihood:

$$\mu_{tu} = \Delta e^{\eta_{tu}}, \qquad \ell_t = \sum_{u=1}^{U} [y_{tu} \log \mu_{tu} - \mu_{tu} - \log(y_{tu}!)].$$

With nuisance x and unit/window history h, the frozen linear predictors are:

$$\eta_{tu}^{B0} = \alpha_u, \qquad \eta_{tu}^{B1} = \alpha_u + \beta_u^{\top} x_t.$$

$$\eta_{tu}^{B2} = \eta_{tu}^{B1} + \sum_{l} a_{ul} h_{t,u,l}.$$

$$\eta_{tu}^{B3} = \eta_{tu}^{B1} + \sum_{v \neq u,\, l} c_{uvl} h_{t,v,l}, \qquad \eta_{tu}^{B4} = \eta_{tu}^{B2} + \sum_{v \neq u,\, l} c_{uvl} h_{t,v,l}.$$

Nominal counts are qB0=U, qB1=38U, qB2=42U, qB3=U[38+4(U-1)], qB4=U(38+4U). For U8:8,304,336,528,560. S10 has41 rather than37 nuisance columns; unit-count cases substitute their actual U. Actual active/free counts are separately retained in all certificates; stored coefficient-vector length is not automatically a nominal count when columns are fixed.

D has a reference-zero state offset, softmax transition rows(reference last logit0) and initial probabilities with the same gauge:

$$g_t(k) = \prod_u \mathrm{Pois}\left(y_{tu}; \Delta e^{\eta_{tu}^{B4} + \delta_{ku}}\right), \qquad \delta_{0u} = 0.$$

$$p_t(y_t) = \sum_k a_t(k) g_t(k), \qquad r_t(k) = \frac{a_t(k) g_t(k)}{\sum_j a_t(j) g_t(j)}, \qquad a_{t+1} = r_t P.$$

Reset a to pi at every epoch; process warm-up without score LL. Conditional score LL equals logL(warm+score)-logL(warm), including the warm-up derivative. Log normalizers are accumulated in float64. Whole-sequence likelihoods are not exponentiated; no posterior smoothing/current-bin plug-in is used. At all delta0, D recovers B4. qD=qB4+(K-1)U+K(K-1)+(K-1), giving571(K2)or584(K3)at U8.

C projects standardized strictly past50-ms microbin rates z through the first r orthonormal DCT-II columns over frozen unit order:

$$R_{v0} = U^{-1/2}, \qquad R_{vj} = \sqrt{2/U} \cos\left(\pi (v + 1/2) j / U\right) \quad (j > 0).$$

$$q_t = \sum_{j=1}^{10} w_j R^{\mathsf{T}} z_{t-j}, \qquad w_j = \frac{e^{-(j-1)0.05/\tau}}{\sum_{i=1}^{10} e^{-(i-1)0.05/\tau}}, \qquad \eta_{tu}^{C} = \eta_{tu}^{B4} + \gamma_u^{\mathsf{T}} \widetilde{q}_t.$$

Here tilde q denotes the separately calibrated state transform. qC=qB4+Ur,568(r1)or576(r2)at U8. Tau is0.1or0.25s and state history truncates at0.5s. No recurrent learned integration, autonomous simulation or estimated projection is present. Added coefficients0 recover B4.

S is an independent per-bin mixture with B4 emissions and constant weights:

$$p_S(y_t) = \sum_{k=1}^{K_S} \pi_k g_t(k), \qquad q_S = q_{B4} + (K_S - 1)U + (K_S - 1).$$

The capacity pair is KS=KD+1, so qS578(K3)or587(K4)at U8. The selected D3/S4 pair has584/587parameters per seed. A same-K HMM whose transition rows all equal pi reproduces the IID mixture mathematically; the registered synthetic identity is separate from this larger static guard. D and S each ensemble three pre-bin joint masses uniformly, maintaining each D seed's own filter. Do not average state coefficients or infer equal effective capacity from the nominal count.

S7. Selection grids, initialization and numerical gates

The objective is normalized by population score bins T:

$$J(\theta) = -\frac{1}{T} \sum_{t \in \text{score}} \ell_t(\theta) + \frac{\lambda}{2} \sum_{i \in \text{nonintercept}} \theta_i^2.$$

Intercepts are unpenalized; all non-intercept free coefficients, including mixture/transition logits and state offsets, are penalized. B0 is the constrained stationary MLE:log(total count/(TDelta)), bounded to[-8,8]; all-zero target optimum-8. Other intercept initialization is log[(training score count+0.5)/(TDelta)], projected to the same box. The0.5pseudocount is only initialization, not a likelihood change. Other linear terms start0. D/S seed11/23/37 offsets are independent Uniform[-0.05,0.05], pi logits0; D's free transition logit in row i and column j is 2 I(i=j) - 2 I(i=K-1), with the last reference column fixed to zero. This is the frozen diagonal preference, not a tuned initializer. No mouse borrows another's fitted coefficients and no Final fit borrows Development coefficients.

Bounds:intercepts[-8,8], linear[-1,1], state offset[-2,2], free logits[-4,4]. B4 penalties and transformations are shared in D/C/S. No predicted rate clipping. All fits use float64. L-BFGS-B maxiter2000, maxfun80000, maxls40, ftol1e-12, gtol1e-6; no retries, alternative solver or budget amendment. B0's closed form is still checked numerically.

For a minimization problem, the frozen outward-gradient projected coordinate is zero only at a lower bound with g>0 or at an upper bound with g<0; otherwise retain g. PG is the infinity norm. Acceptance is PG<=0.001. Independent real endpoint AD and every central finite-difference step1e-4,1e-5,1e-6 must satisfy max coordinate |g-gref|/max(1,|g|,|gref|)<=1e-5; no favorable-step selection. Synthetic analytic/AD error is<=1e-8. At bounds, finite differences compare the mathematical extension's unconstrained derivative before PG is applied. A finite-difference explanation does not clear failure of its original gate. Every optimizer termination, failed seed and partial start remains recorded.

Development grid:ridge0.01/0.1/1; D K2/3 with3seeds; C r1/2 andtau0.1/0.25; S K3/4 with3seeds. Per mouse:1B0+12otherbaseline+18D+12C+18S=61starts. The corrected ten-mouse main total was610. Each candidate required all fixed animals/starts; B selection used equal-mouse LL/Q. Ties within1e-6 favored fewer nominal parameters, stronger ridge, smaller K/r, smaller tau, then lexical ID. S was selected across ridge only at KD+1. No E or Final choice altered that grid.

Selected policies:B0; B1_l0.01; B2_l0.01; B3_l1.0; B4_l0.1; C_r1_t0.1_l0.1; D_k3_l0.01; S_k4_l0.01. Final used12main fits per admitted mouse and no search. The complete table of37 seed-grouped candidates representing61 optimizer starts per mouse follows, including unselected configurations and nominal counts; the immutable original table is copied byte-for-byte.

Configuration	B LL/unit-bin	Nominal q	Selected
B0	-0.9915065395	8	True
B1_l0.01	-0.8989816559	304	True
B1_l0.1	-0.8991664954	304	False
B1_l1.0	-0.9001828796	304	False
B2_l0.01	-0.8543851683	336	True
B2_l0.1	-0.8547361175	336	False
B2_l1.0	-0.8708143614	336	False
B3_l0.01	-0.8996660688	528	False
B3_l0.1	-0.8886034998	528	False
B3_l1.0	-0.8871836557	528	True
B4_l0.01	-0.8572680559	560	False
B4_l0.1	-0.8503982647	560	True
B4_l1.0	-0.8634147755	560	False
D_k2_l0.01	-0.8437139809	571	False
D_k2_l0.1	-0.8481113974	571	False
D_k2_l1.0	-0.8634147756	571	False
D_k3_l0.01	-0.8410101819	584	True
D_k3_l0.1	-0.845886661	584	False

Configuration	B LL/unit-bin	Nominal q	Selected
D_k3_l1.0	-0.8634147719	584	False
C_r1_t0.1_l0.01	-0.8573305244	568	False
C_r1_t0.1_l0.1	-0.850480291	568	True
C_r1_t0.1_l1.0	-0.8629618939	568	False
C_r1_t0.25_l0.01	-0.8573687475	568	False
C_r1_t0.25_l0.1	-0.8504835277	568	False
C_r1_t0.25_l1.0	-0.8630530237	568	False
C_r2_t0.1_l0.01	-0.8574176475	576	False
C_r2_t0.1_l0.1	-0.8505600675	576	False
C_r2_t0.1_l1.0	-0.8625424985	576	False
C_r2_t0.25_l0.01	-0.8574485586	576	False
C_r2_t0.25_l0.1	-0.8505151442	576	False
C_r2_t0.25_l1.0	-0.8625945622	576	False
S_k3_l0.01	-0.8407447671	578	False
S_k3_l0.1	-0.8458782343	578	False
S_k3_l1.0	-0.8634147785	578	False
S_k4_l0.01	-0.8391156589	587	True
S_k4_l0.1	-0.844934764	587	False
S_k4_l1.0	-0.8634147785	587	False

S8. Complete Final numerical acceptance

All72main and320sensitivity C fits passed; no retries occurred. There are392endpoint certificates. Full path, source SHA, status, active/free count, stored coefficient-vector length, PG, AD, max of the three FD errors and iterations are supplied in `tables/FINAL_NUMERICAL_CERTIFICATES.csv`; original main/sensitivity attempts are also copied unchanged. The following table includes every completed endpoint, with compact identifiers. Configuration names identify original seed/ridge/rank; baseline sensitivity policies use their unchanged selected ridges. Scientific positive-claim status remains independent of numerical success.

C fit group	Endpoints	Largest PG	Largest AD	Largest FD
Main	72	1.329192e-05	2.37113e-14	2.040363e-08
Sensitivity	320	1.239118e-05	2.608161e-14	3.215293e-08

Mouse/case/configuration	Free q	PG	AD error	FD max	Iter
DY_011/PRIMARY/B0	8	3.823e-15	3.777e-15	3.473e-09	0
DY_011/PRIMARY/B1_l0.01	304	8.105e-07	7.75e-17	8.452e-09	272
DY_011/PRIMARY/B2_l0.01	336	9.674e-07	6.072e-17	8.424e-09	283
DY_011/PRIMARY/B3_l1.0	528	1.244e-06	1.527e-16	8.458e-09	71
DY_011/PRIMARY/B4_l0.1	560	1.085e-06	6.072e-17	8.4e-09	115

Mouse/case/configuration	Free q	PG	AD error	FD max	Iter
DY_011/PRIMARY/C_r1_t0.1_l0.1	568	7.559e-07	7.633e-17	8.405e-09	117
DY_011/PRIMARY/D_k3_l0.01_s11	584	9.38e-07	7.878e-16	8.385e-09	354
DY_011/PRIMARY/D_k3_l0.01_s23	584	7.706e-07	7.677e-16	8.38e-09	401
DY_011/PRIMARY/D_k3_l0.01_s37	584	9.675e-07	4.881e-16	8.372e-09	407
DY_011/PRIMARY/S_k4_l0.01_s11	587	8.713e-07	1.087e-16	8.374e-09	396
DY_011/PRIMARY/S_k4_l0.01_s23	587	9.663e-07	1.61e-16	8.376e-09	342
DY_011/PRIMARY/S_k4_l0.01_s37	587	8.783e-07	4.171e-16	8.373e-09	361
DY_014/PRIMARY/B0	8	1.846e-14	1.879e-14	3.97e-09	0
DY_014/PRIMARY/B1_l0.01	304	1.861e-06	4.341e-16	1.118e-08	261
DY_014/PRIMARY/B2_l0.01	336	3.574e-06	2.22e-16	1.118e-08	267
DY_014/PRIMARY/B3_l1.0	528	2.658e-06	3.782e-16	1.09e-08	47
DY_014/PRIMARY/B4_l0.1	560	1.329e-06	2.884e-16	1.115e-08	108
DY_014/PRIMARY/C_r1_t0.1_l0.1	568	4.558e-06	1.995e-16	1.116e-08	108
DY_014/PRIMARY/D_k3_l0.01_s11	584	2.617e-06	3.894e-15	1.038e-08	647
DY_014/PRIMARY/D_k3_l0.01_s23	584	2.537e-06	6.281e-15	1.045e-08	400
DY_014/PRIMARY/D_k3_l0.01_s37	584	1.483e-06	3.699e-15	1.045e-08	403
DY_014/PRIMARY/S_k4_l0.01_s11	587	2.047e-06	4.605e-16	1.059e-08	347
DY_014/PRIMARY/S_k4_l0.01_s23	587	2.208e-06	3.851e-16	1.058e-08	413
DY_014/PRIMARY/S_k4_l0.01_s37	587	6.462e-06	3.669e-16	1.059e-08	452
NR_0027/PRIMARY/B0	8	7.219e-15	7.033e-15	1.07e-09	0
NR_0027/PRIMARY/B1_l0.01	304	7.069e-07	8.929e-17	3.033e-09	128
NR_0027/PRIMARY/B2_l0.01	336	8.927e-07	7.86e-17	3.026e-09	132
NR_0027/PRIMARY/B3_l1.0	528	7.8e-07	2.776e-16	3.048e-09	60
NR_0027/PRIMARY/B4_l0.1	560	9.4e-07	1.006e-16	3.044e-09	70
NR_0027/PRIMARY/C_r1_t0.1_l0.1	568	9.464e-07	1.362e-16	3.042e-09	82
NR_0027/PRIMARY/D_k3_l0.01_s11	584	9.542e-07	5.742e-16	2.977e-09	219
NR_0027/PRIMARY/D_k3_l0.01_s23	584	9.993e-07	3.162e-16	2.987e-09	179
NR_0027/PRIMARY/D_k3_l0.01_s37	584	8.267e-07	5.612e-16	2.985e-09	165
NR_0027/PRIMARY/S_k4_l0.01_s11	587	9.068e-07	7.3e-16	2.987e-09	175
NR_0027/PRIMARY/S_k4_l0.01_s23	587	9.322e-07	4.229e-16	2.987e-09	189
NR_0027/PRIMARY/S_k4_l0.01_s37	587	8.48e-07	5.046e-16	2.988e-09	167
UCLA037/PRIMARY/B0	8	1.218e-14	1.225e-14	5.818e-09	0
UCLA037/PRIMARY/B1_l0.01	304	1.329e-05	2.333e-16	1.971e-08	266
UCLA037/PRIMARY/B2_l0.01	336	2.611e-06	2.793e-16	1.97e-08	283
UCLA037/PRIMARY/B3_l1.0	528	7.557e-07	3.192e-16	2.04e-08	40
UCLA037/PRIMARY/B4_l0.1	560	6.572e-06	2.307e-16	1.985e-08	107
UCLA037/PRIMARY/C_r1_t0.1_l0.1	568	1.662e-06	2.051e-16	1.985e-08	107
UCLA037/PRIMARY/D_k3_l0.01_s11	584	1.503e-06	6.807e-15	1.961e-08	408
UCLA037/PRIMARY/D_k3_l0.01_s23	584	1.092e-05	4.341e-15	1.959e-08	599

Mouse/case/configuration	Free q	PG	AD error	FD max	Iter
UCLA037/PRIMARY/D_k3_l0.01_s37	584	1.415e-06	1.352e-14	1.959e-08	569
UCLA037/PRIMARY/S_k4_l0.01_s11	587	7.834e-06	3.771e-16	1.963e-08	352
UCLA037/PRIMARY/S_k4_l0.01_s23	587	3.685e-06	4.369e-16	1.964e-08	340
UCLA037/PRIMARY/S_k4_l0.01_s37	587	2.413e-06	4.778e-16	1.964e-08	347
ZFM-01576/PRIMARY/B0	8	2.326e-14	2.371e-14	1.878e-09	0
ZFM-01576/PRIMARY/B1_l0.01	304	9.506e-07	9.845e-17	5.151e-09	163
ZFM-01576/PRIMARY/B2_l0.01	336	1.919e-06	1.082e-16	5.149e-09	168
ZFM-01576/PRIMARY/B3_l1.0	528	1.141e-06	3.747e-16	4.809e-09	39
ZFM-01576/PRIMARY/B4_l0.1	560	7.655e-07	1.83e-16	5.101e-09	66
ZFM-01576/PRIMARY/C_r1_t0.1_l0.1	568	6.054e-07	1.639e-16	5.109e-09	68
ZFM-01576/PRIMARY/D_k3_l0.01_s11	584	8.133e-07	1.531e-15	4.841e-09	254
ZFM-01576/PRIMARY/D_k3_l0.01_s23	584	9.593e-07	4.593e-16	4.847e-09	251
ZFM-01576/PRIMARY/D_k3_l0.01_s37	584	9.952e-07	5.277e-16	4.832e-09	294
ZFM-01576/PRIMARY/S_k4_l0.01_s11	587	9.269e-07	2.977e-16	4.867e-09	263
ZFM-01576/PRIMARY/S_k4_l0.01_s23	587	3.417e-06	1.141e-15	4.867e-09	320
ZFM-01576/PRIMARY/S_k4_l0.01_s37	587	9.659e-07	7.594e-16	4.869e-09	300
ZFM-01936/PRIMARY/B0	8	4.363e-15	4.777e-15	2.567e-09	0
ZFM-01936/PRIMARY/B1_l0.01	304	1.345e-06	1.002e-16	7.28e-09	192
ZFM-01936/PRIMARY/B2_l0.01	336	9.31e-07	9.584e-17	7.284e-09	194
ZFM-01936/PRIMARY/B3_l1.0	528	1.599e-06	1.874e-16	6.665e-09	37
ZFM-01936/PRIMARY/B4_l0.1	560	7.346e-07	1.37e-16	7e-09	71
ZFM-01936/PRIMARY/C_r1_t0.1_l0.1	568	8.077e-07	9.714e-17	7.002e-09	71
ZFM-01936/PRIMARY/D_k3_l0.01_s11	584	8.762e-07	5.333e-15	7.288e-09	218
ZFM-01936/PRIMARY/D_k3_l0.01_s23	584	8.273e-07	8.389e-15	7.289e-09	216
ZFM-01936/PRIMARY/D_k3_l0.01_s37	584	3.193e-06	9.527e-15	7.289e-09	209
ZFM-01936/PRIMARY/S_k4_l0.01_s11	587	8.982e-07	7.761e-16	7.28e-09	203
ZFM-01936/PRIMARY/S_k4_l0.01_s23	587	9.716e-07	5.593e-16	7.284e-09	204
ZFM-01936/PRIMARY/S_k4_l0.01_s37	587	2.009e-06	4.869e-16	7.28e-09	204
DY_011/S01/B0	5	3.823e-15	3.777e-15	3.482e-09	0
DY_011/S01/B1	190	9.702e-07	7.566e-17	8.459e-09	176
DY_011/S01/B2	210	1.954e-06	6.072e-17	8.425e-09	186
DY_011/S01/B3	270	6.787e-07	1.388e-16	8.454e-09	41
DY_011/S01/B4	290	1.244e-06	6.105e-17	8.409e-09	70
DY_011/S02/B0	12	3.823e-15	3.777e-15	4.361e-09	0
DY_011/S02/B1	456	1.652e-06	8.804e-17	1.044e-08	368
DY_011/S02/B2	504	1.07e-05	1.367e-16	1.045e-08	364
DY_011/S02/B3	984	2.626e-06	2.082e-16	1.072e-08	97
DY_011/S02/B4	1032	1.579e-06	1.791e-16	1.049e-08	176
DY_011/S04/B0	8	3.994e-15	4.082e-15	6.954e-09	0

Mouse/case/configuration	Free q	PG	AD error	FD max	Iter
DY_011/S04/B1	304	8.208e-06	1.644e-16	1.548e-08	328
DY_011/S04/B2	336	1.239e-05	1.778e-16	1.545e-08	356
DY_011/S04/B3	528	5.406e-06	1.943e-16	1.564e-08	79
DY_011/S04/B4	560	1.146e-06	9.714e-17	1.551e-08	157
DY_011/S05/B0	8	4.914e-15	5.083e-15	3.446e-09	0
DY_011/S05/B1	304	8.701e-07	1.426e-16	7.951e-09	264
DY_011/S05/B2	336	7.481e-07	1.66e-16	7.939e-09	221
DY_011/S05/B3	528	2.511e-06	4.163e-16	7.956e-09	66
DY_011/S05/B4	560	1.609e-06	1.659e-16	7.922e-09	95
DY_011/S06/B0	8	3.823e-15	3.777e-15	3.473e-09	0
DY_011/S06/B1	304	8.105e-07	7.75e-17	8.452e-09	272
DY_011/S06/B2	336	9.674e-07	6.072e-17	8.424e-09	283
DY_011/S06/B3	528	1.244e-06	1.527e-16	8.458e-09	71
DY_011/S06/B4	560	1.085e-06	6.072e-17	8.4e-09	115
DY_011/S07/B0	8	9.901e-16	9.002e-16	4.441e-10	0
DY_011/S07/B1	304	9.451e-07	3.86e-17	8.663e-10	99
DY_011/S07/B2	336	8.51e-07	2.819e-17	1.233e-09	95
DY_011/S07/B3	528	7.97e-07	1.318e-16	7.133e-10	35
DY_011/S07/B4	560	9.077e-07	7.286e-17	9.576e-10	46
DY_011/S08/B0	8	3.823e-15	3.777e-15	3.473e-09	0
DY_011/S08/B1	304	8.105e-07	7.75e-17	8.452e-09	272
DY_011/S08/B2	336	9.674e-07	6.072e-17	8.424e-09	283
DY_011/S08/B3	528	1.244e-06	1.527e-16	8.458e-09	71
DY_011/S08/B4	560	1.085e-06	6.072e-17	8.4e-09	115
DY_011/S09/B0	8	3.823e-15	3.777e-15	3.473e-09	0
DY_011/S09/B1	304	8.105e-07	7.75e-17	8.452e-09	272
DY_011/S09/B2	336	9.674e-07	6.072e-17	8.424e-09	283
DY_011/S09/B3	528	1.244e-06	1.527e-16	8.458e-09	71
DY_011/S09/B4	560	1.085e-06	6.072e-17	8.4e-09	115
DY_011/S10/B0	8	3.823e-15	3.777e-15	3.473e-09	0
DY_011/S10/B1	336	7.803e-07	6.559e-17	8.458e-09	310
DY_011/S10/B2	368	6.901e-06	5.421e-17	8.42e-09	292
DY_011/S10/B3	560	1.273e-06	2.22e-16	8.458e-09	71
DY_011/S10/B4	592	7.377e-07	1.076e-16	8.406e-09	143
DY_011/S11/B0	5	3.823e-15	3.777e-15	3.482e-09	0
DY_011/S11/B1	190	9.702e-07	7.566e-17	8.459e-09	176
DY_011/S11/B2	210	1.954e-06	6.072e-17	8.425e-09	186
DY_011/S11/B3	270	6.787e-07	1.388e-16	8.454e-09	41
DY_011/S11/B4	290	1.244e-06	6.105e-17	8.409e-09	70

Mouse/case/configuration	Free q	PG	AD error	FD max	Iter
DY_014/S01/B0	5	1.846e-14	1.879e-14	3.961e-09	0
DY_014/S01/B1	190	1.382e-06	3.888e-16	1.118e-08	255
DY_014/S01/B2	210	1.351e-06	2.558e-16	1.116e-08	265
DY_014/S01/B3	270	1.249e-06	4.441e-16	1.089e-08	37
DY_014/S01/B4	290	3.156e-06	2.285e-16	1.115e-08	104
DY_014/S02/B0	12	1.846e-14	1.879e-14	3.961e-09	0
DY_014/S02/B1	456	5.328e-06	4.704e-16	1.117e-08	262
DY_014/S02/B2	504	3.343e-06	1.874e-16	1.116e-08	267
DY_014/S02/B3	984	1.913e-06	4.545e-16	1.089e-08	52
DY_014/S02/B4	1032	4.03e-06	2.68e-16	1.113e-08	112
DY_014/S03/B0	20	1.846e-14	1.879e-14	3.944e-09	0
DY_014/S03/B1	760	6.179e-06	3.205e-16	1.119e-08	259
DY_014/S03/B2	840	9.001e-06	2.955e-16	1.116e-08	268
DY_014/S03/B3	2280	7.097e-06	4.614e-16	1.086e-08	103
DY_014/S03/B4	2360	2.099e-06	2.238e-16	1.116e-08	152
DY_014/S04/B0	8	6.477e-15	6.703e-15	7.923e-09	0
DY_014/S04/B1	304	9.034e-06	5.972e-16	2.314e-08	364
DY_014/S04/B2	336	7.064e-06	2.732e-16	2.314e-08	373
DY_014/S04/B3	528	3.371e-06	5.204e-16	2.237e-08	51
DY_014/S04/B4	560	2.172e-06	4.137e-16	2.302e-08	148
DY_014/S05/B0	8	1.446e-14	1.46e-14	4.006e-09	0
DY_014/S05/B1	304	9.282e-06	4.189e-16	1.082e-08	239
DY_014/S05/B2	336	5.944e-06	2.591e-16	1.079e-08	262
DY_014/S05/B3	528	2.48e-06	2.914e-16	1.068e-08	42
DY_014/S05/B4	560	1.765e-06	3.053e-16	1.072e-08	110
DY_014/S06/B0	8	1.846e-14	1.879e-14	3.97e-09	0
DY_014/S06/B1	304	1.861e-06	4.341e-16	1.118e-08	261
DY_014/S06/B2	336	3.574e-06	2.22e-16	1.118e-08	267
DY_014/S06/B3	528	2.658e-06	3.782e-16	1.09e-08	47
DY_014/S06/B4	560	1.329e-06	2.884e-16	1.115e-08	108
DY_014/S07/B0	8	5.595e-16	5.842e-16	4.441e-10	0
DY_014/S07/B1	304	9.521e-07	4.571e-17	1.019e-09	77
DY_014/S07/B2	336	7.887e-07	5.317e-17	1.022e-09	79
DY_014/S07/B3	528	7.85e-07	7.633e-17	8.94e-10	30
DY_014/S07/B4	560	7.933e-07	4.597e-17	9.971e-10	35
DY_014/S08/B0	8	1.846e-14	1.879e-14	3.97e-09	0
DY_014/S08/B1	304	1.861e-06	4.341e-16	1.118e-08	261
DY_014/S08/B2	336	3.574e-06	2.22e-16	1.118e-08	267
DY_014/S08/B3	528	2.658e-06	3.782e-16	1.09e-08	47

Mouse/case/configuration	Free q	PG	AD error	FD max	Iter
DY_014/S08/B4	560	1.329e-06	2.884e-16	1.115e-08	108
DY_014/S09/B0	8	1.846e-14	1.879e-14	3.97e-09	0
DY_014/S09/B1	304	1.861e-06	4.341e-16	1.118e-08	261
DY_014/S09/B2	336	3.574e-06	2.22e-16	1.118e-08	267
DY_014/S09/B3	528	2.658e-06	3.782e-16	1.09e-08	47
DY_014/S09/B4	560	1.329e-06	2.884e-16	1.115e-08	108
DY_014/S10/B0	8	1.846e-14	1.879e-14	3.97e-09	0
DY_014/S10/B1	336	1.911e-06	4.163e-16	1.117e-08	267
DY_014/S10/B2	368	6.037e-06	2.099e-16	1.117e-08	263
DY_014/S10/B3	560	4.482e-06	3.608e-16	1.09e-08	46
DY_014/S10/B4	592	1.899e-06	3.665e-16	1.115e-08	103
DY_014/S11/B0	5	1.846e-14	1.879e-14	3.961e-09	0
DY_014/S11/B1	190	1.382e-06	3.888e-16	1.118e-08	255
DY_014/S11/B2	210	1.351e-06	2.558e-16	1.116e-08	265
DY_014/S11/B3	270	1.249e-06	4.441e-16	1.089e-08	37
DY_014/S11/B4	290	3.156e-06	2.285e-16	1.115e-08	104
NR_0027/S01/B0	5	7.219e-15	7.033e-15	1.07e-09	0
NR_0027/S01/B1	190	9.249e-07	1.075e-16	3.031e-09	129
NR_0027/S01/B2	210	8.792e-07	1.134e-16	3.029e-09	126
NR_0027/S01/B3	270	6.593e-07	1.735e-16	3.059e-09	36
NR_0027/S01/B4	290	6.196e-07	1.232e-16	3.046e-09	54
NR_0027/S02/B0	12	7.219e-15	7.033e-15	1.07e-09	0
NR_0027/S02/B1	456	7.15e-07	9.498e-17	3.037e-09	124
NR_0027/S02/B2	504	6.64e-07	1.119e-16	3.031e-09	131
NR_0027/S02/B3	984	9.845e-07	2.22e-16	3.485e-09	63
NR_0027/S02/B4	1032	3.659e-06	1.132e-16	3.811e-09	86
NR_0027/S04/B0	8	1.004e-14	9.645e-15	2.145e-09	0
NR_0027/S04/B1	304	3.063e-06	9.867e-17	6.827e-09	189
NR_0027/S04/B2	336	7.197e-07	2.086e-16	6.841e-09	191
NR_0027/S04/B3	528	1.134e-06	4.025e-16	6.946e-09	62
NR_0027/S04/B4	560	9.696e-07	1.18e-16	6.875e-09	83
NR_0027/S05/B0	8	1.316e-14	1.275e-14	1.075e-09	0
NR_0027/S05/B1	304	8.879e-07	1.475e-16	3.132e-09	130
NR_0027/S05/B2	336	8.726e-07	8.11e-17	3.129e-09	131
NR_0027/S05/B3	528	1.116e-06	2.429e-16	3.204e-09	43
NR_0027/S05/B4	560	8.286e-07	8.5e-17	3.154e-09	51
NR_0027/S06/B0	8	7.219e-15	7.033e-15	1.07e-09	0
NR_0027/S06/B1	304	7.069e-07	8.929e-17	3.033e-09	128
NR_0027/S06/B2	336	8.927e-07	7.86e-17	3.026e-09	132

Mouse/case/configuration	Free q	PG	AD error	FD max	Iter
NR_0027/S06/B3	528	7.8e-07	2.776e-16	3.048e-09	60
NR_0027/S06/B4	560	9.4e-07	1.006e-16	3.044e-09	70
NR_0027/S07/B0	8	3.843e-15	3.974e-15	2.687e-10	0
NR_0027/S07/B1	304	9.154e-07	4.38e-17	9.842e-10	74
NR_0027/S07/B2	336	9.108e-07	6.137e-17	1.133e-09	81
NR_0027/S07/B3	528	9.312e-07	2.009e-16	7.569e-10	40
NR_0027/S07/B4	560	9.703e-07	6.072e-17	9.78e-10	43
NR_0027/S08/B0	8	7.219e-15	7.033e-15	1.07e-09	0
NR_0027/S08/B1	304	7.069e-07	8.929e-17	3.033e-09	128
NR_0027/S08/B2	336	8.927e-07	7.86e-17	3.026e-09	132
NR_0027/S08/B3	528	7.8e-07	2.776e-16	3.048e-09	60
NR_0027/S08/B4	560	9.4e-07	1.006e-16	3.044e-09	70
NR_0027/S09/B0	8	7.219e-15	7.033e-15	1.07e-09	0
NR_0027/S09/B1	304	7.069e-07	8.929e-17	3.033e-09	128
NR_0027/S09/B2	336	8.927e-07	7.86e-17	3.026e-09	132
NR_0027/S09/B3	528	7.8e-07	2.776e-16	3.048e-09	60
NR_0027/S09/B4	560	9.4e-07	1.006e-16	3.044e-09	70
NR_0027/S10/B0	8	7.219e-15	7.033e-15	1.07e-09	0
NR_0027/S10/B1	336	9.249e-07	1.145e-16	3.032e-09	134
NR_0027/S10/B2	368	8.79e-07	1.42e-16	3.027e-09	142
NR_0027/S10/B3	560	7.64e-07	2.082e-16	3.056e-09	64
NR_0027/S10/B4	592	9.641e-07	8.63e-17	3.041e-09	85
NR_0027/S11/B0	5	7.219e-15	7.033e-15	1.07e-09	0
NR_0027/S11/B1	190	9.249e-07	1.075e-16	3.031e-09	129
NR_0027/S11/B2	210	8.792e-07	1.134e-16	3.029e-09	126
NR_0027/S11/B3	270	6.593e-07	1.735e-16	3.059e-09	36
NR_0027/S11/B4	290	6.196e-07	1.232e-16	3.046e-09	54
UCLA037/S01/B0	5	1.218e-14	1.225e-14	5.826e-09	0
UCLA037/S01/B1	190	8.397e-06	2.713e-16	1.969e-08	262
UCLA037/S01/B2	210	1.362e-06	2.072e-16	1.97e-08	279
UCLA037/S01/B3	270	2.826e-06	5.482e-16	2.036e-08	37
UCLA037/S01/B4	290	7.966e-06	2.472e-16	1.983e-08	100
UCLA037/S02/B0	12	1.218e-14	1.225e-14	5.809e-09	0
UCLA037/S02/B1	456	1.084e-05	2.535e-16	1.969e-08	266
UCLA037/S02/B2	504	7.909e-06	2.407e-16	1.968e-08	279
UCLA037/S02/B3	984	2.748e-06	2.776e-16	2.044e-08	49
UCLA037/S02/B4	1032	1.327e-06	2.351e-16	1.984e-08	108
UCLA037/S03/B0	20	1.218e-14	1.225e-14	5.809e-09	0
UCLA037/S03/B1	760	2.671e-06	3.157e-16	1.974e-08	282

Mouse/case/configuration	Free q	PG	AD error	FD max	Iter
UCLA037/S03/B2	840	2.208e-06	2.55e-16	1.968e-08	288
UCLA037/S03/B3	2280	2.119e-06	2.776e-16	2.047e-08	60
UCLA037/S03/B4	2360	1.709e-06	2.463e-16	1.985e-08	123
UCLA037/S04/B0	8	7.899e-15	8.03e-15	1.164e-08	0
UCLA037/S04/B1	304	6.1e-06	3.502e-16	3.215e-08	383
UCLA037/S04/B2	336	2.532e-06	3.733e-16	3.213e-08	396
UCLA037/S04/B3	528	2.753e-06	5.482e-16	3.192e-08	51
UCLA037/S04/B4	560	2.896e-06	3.779e-16	3.211e-08	148
UCLA037/S05/B0	8	1.616e-14	1.649e-14	5.871e-09	0
UCLA037/S05/B1	304	1.007e-05	3.036e-16	1.974e-08	255
UCLA037/S05/B2	336	1.636e-06	2.136e-16	1.973e-08	281
UCLA037/S05/B3	528	3.116e-06	3.261e-16	2.042e-08	39
UCLA037/S05/B4	560	1.481e-06	2.661e-16	1.986e-08	104
UCLA037/S06/B0	8	1.218e-14	1.225e-14	4.947e-09	0
UCLA037/S06/B1	304	5.634e-06	1.921e-16	1.732e-08	244
UCLA037/S06/B2	336	4.879e-06	2.011e-16	1.731e-08	261
UCLA037/S06/B3	528	9.282e-06	2.706e-16	1.697e-08	37
UCLA037/S06/B4	560	1.611e-06	2.073e-16	1.731e-08	101
UCLA037/S07/B0	8	5.31e-16	5.754e-16	4.441e-10	0
UCLA037/S07/B1	304	7.968e-07	2.976e-17	1.006e-09	67
UCLA037/S07/B2	336	6.819e-07	1.952e-17	1.006e-09	64
UCLA037/S07/B3	528	9.237e-07	7.286e-17	8.706e-10	21
UCLA037/S07/B4	560	4.443e-07	3.209e-17	9.868e-10	27
UCLA037/S08/B0	8	5.259e-15	5.252e-15	5.8e-09	0
UCLA037/S08/B1	304	2.222e-06	2.323e-16	1.962e-08	274
UCLA037/S08/B2	336	8.455e-06	2.257e-16	1.96e-08	273
UCLA037/S08/B3	528	2.53e-06	4.103e-16	2.032e-08	40
UCLA037/S08/B4	560	1.585e-06	2.086e-16	1.974e-08	106
UCLA037/S09/B0	8	1.18e-14	1.112e-14	5.782e-09	0
UCLA037/S09/B1	304	1.639e-06	3.831e-16	1.956e-08	275
UCLA037/S09/B2	336	1.778e-06	2.608e-16	1.956e-08	288
UCLA037/S09/B3	528	2.48e-06	4.302e-16	2.029e-08	39
UCLA037/S09/B4	560	4.441e-06	2.741e-16	1.971e-08	110
UCLA037/S10/B0	8	1.218e-14	1.225e-14	5.818e-09	0
UCLA037/S10/B1	336	1.978e-06	1.649e-16	1.971e-08	292
UCLA037/S10/B2	368	2.413e-06	2.248e-16	1.969e-08	294
UCLA037/S10/B3	560	4.046e-06	3.053e-16	2.042e-08	41
UCLA037/S10/B4	592	1.577e-06	2.172e-16	1.983e-08	114
UCLA037/S11/B0	5	1.218e-14	1.225e-14	5.826e-09	0

Mouse/case/configuration	Free q	PG	AD error	FD max	Iter
UCLA037/S11/B1	190	8.397e-06	2.713e-16	1.969e-08	262
UCLA037/S11/B2	210	1.362e-06	2.072e-16	1.97e-08	279
UCLA037/S11/B3	270	2.826e-06	5.482e-16	2.036e-08	37
UCLA037/S11/B4	290	7.966e-06	2.472e-16	1.983e-08	100
ZFM-01576/S01/B0	5	2.326e-14	2.371e-14	1.883e-09	0
ZFM-01576/S01/B1	190	3.389e-06	1.028e-16	5.156e-09	152
ZFM-01576/S01/B2	210	8.536e-07	1.379e-16	5.147e-09	159
ZFM-01576/S01/B3	270	1.082e-06	3.053e-16	4.779e-09	32
ZFM-01576/S01/B4	290	8.64e-07	1.631e-16	5.098e-09	62
ZFM-01576/S02/B0	12	2.326e-14	2.371e-14	1.883e-09	0
ZFM-01576/S02/B1	456	9.616e-07	6.592e-17	5.163e-09	166
ZFM-01576/S02/B2	504	9.834e-07	1.193e-16	5.161e-09	164
ZFM-01576/S02/B3	984	1.755e-06	3.886e-16	4.817e-09	62
ZFM-01576/S02/B4	1032	9.005e-07	1.7e-16	5.099e-09	81
ZFM-01576/S03/B0	20	2.326e-14	2.371e-14	1.874e-09	0
ZFM-01576/S03/B1	760	3.863e-06	1.119e-16	5.156e-09	172
ZFM-01576/S03/B2	840	7.711e-07	9.649e-17	5.15e-09	182
ZFM-01576/S03/B3	2280	1.849e-06	4.58e-16	4.826e-09	102
ZFM-01576/S03/B4	2360	8.506e-07	1.475e-16	5.098e-09	142
ZFM-01576/S04/B0	8	1.676e-14	1.647e-14	3.766e-09	0
ZFM-01576/S04/B1	304	1.178e-06	9.064e-17	9.927e-09	228
ZFM-01576/S04/B2	336	7.178e-06	1.186e-16	9.928e-09	231
ZFM-01576/S04/B3	528	3.34e-06	6.523e-16	9.669e-09	43
ZFM-01576/S04/B4	560	8.425e-07	1.9e-16	9.912e-09	93
ZFM-01576/S05/B0	8	2.595e-14	2.608e-14	1.878e-09	0
ZFM-01576/S05/B1	304	1.143e-06	1.606e-16	5.152e-09	176
ZFM-01576/S05/B2	336	8.973e-07	1.72e-16	5.153e-09	176
ZFM-01576/S05/B3	528	1.716e-06	2.942e-16	4.824e-09	42
ZFM-01576/S05/B4	560	8.818e-07	3.747e-16	5.109e-09	68
ZFM-01576/S06/B0	8	2.326e-14	2.371e-14	1.878e-09	0
ZFM-01576/S06/B1	304	9.506e-07	9.845e-17	5.151e-09	163
ZFM-01576/S06/B2	336	1.919e-06	1.082e-16	5.149e-09	168
ZFM-01576/S06/B3	528	1.141e-06	3.747e-16	4.809e-09	39
ZFM-01576/S06/B4	560	7.655e-07	1.83e-16	5.101e-09	66
ZFM-01576/S07/B0	8	4.478e-15	4.456e-15	4.441e-10	0
ZFM-01576/S07/B1	304	9.623e-07	5.204e-17	9.561e-10	65
ZFM-01576/S07/B2	336	8.764e-07	3.61e-17	9.69e-10	64
ZFM-01576/S07/B3	528	6.66e-07	1.457e-16	9.662e-10	20
ZFM-01576/S07/B4	560	6.229e-07	7.806e-17	9.232e-10	28

Mouse/case/configuration	Free q	PG	AD error	FD max	Iter
ZFM-01576/S08/B0	8	2.326e-14	2.371e-14	1.878e-09	0
ZFM-01576/S08/B1	304	9.506e-07	9.845e-17	5.151e-09	163
ZFM-01576/S08/B2	336	1.919e-06	1.082e-16	5.149e-09	168
ZFM-01576/S08/B3	528	1.141e-06	3.747e-16	4.809e-09	39
ZFM-01576/S08/B4	560	7.655e-07	1.83e-16	5.101e-09	66
ZFM-01576/S09/B0	8	2.326e-14	2.371e-14	1.878e-09	0
ZFM-01576/S09/B1	304	9.506e-07	9.845e-17	5.151e-09	163
ZFM-01576/S09/B2	336	1.919e-06	1.082e-16	5.149e-09	168
ZFM-01576/S09/B3	528	1.141e-06	3.747e-16	4.809e-09	39
ZFM-01576/S09/B4	560	7.655e-07	1.83e-16	5.101e-09	66
ZFM-01576/S10/B0	8	2.326e-14	2.371e-14	1.878e-09	0
ZFM-01576/S10/B1	336	2.593e-06	7.264e-17	5.154e-09	171
ZFM-01576/S10/B2	368	9.985e-07	1.529e-16	5.152e-09	174
ZFM-01576/S10/B3	560	2.557e-06	3.608e-16	4.813e-09	42
ZFM-01576/S10/B4	592	7.687e-07	1.344e-16	5.102e-09	68
ZFM-01576/S11/B0	5	2.326e-14	2.371e-14	1.883e-09	0
ZFM-01576/S11/B1	190	3.389e-06	1.028e-16	5.156e-09	152
ZFM-01576/S11/B2	210	8.536e-07	1.379e-16	5.147e-09	159
ZFM-01576/S11/B3	270	1.082e-06	3.053e-16	4.779e-09	32
ZFM-01576/S11/B4	290	8.64e-07	1.631e-16	5.098e-09	62
ZFM-01936/S01/B0	5	4.363e-15	4.777e-15	2.554e-09	0
ZFM-01936/S01/B1	190	8.909e-07	8.544e-17	6.783e-09	182
ZFM-01936/S01/B2	210	9.648e-07	7.644e-17	6.785e-09	177
ZFM-01936/S01/B3	270	8.103e-07	3.088e-16	6.621e-09	29
ZFM-01936/S01/B4	290	8.901e-07	1.301e-16	6.758e-09	67
ZFM-01936/S02/B0	12	6.741e-15	6.529e-15	2.549e-09	0
ZFM-01936/S02/B1	456	1.124e-06	1.35e-16	7.301e-09	193
ZFM-01936/S02/B2	504	6.123e-06	1.041e-16	7.299e-09	188
ZFM-01936/S02/B3	984	3.133e-06	3.712e-16	6.712e-09	43
ZFM-01936/S02/B4	1032	9.278e-07	1.509e-16	7.018e-09	73
ZFM-01936/S03/B0	20	6.741e-15	6.529e-15	3.553e-09	0
ZFM-01936/S03/B1	760	4.294e-06	2.16e-16	7.285e-09	201
ZFM-01936/S03/B2	840	1.208e-06	1.182e-16	7.293e-09	203
ZFM-01936/S03/B3	2280	1.864e-06	2.776e-16	6.8e-09	67
ZFM-01936/S03/B4	2360	1.334e-06	1.284e-16	6.997e-09	96
ZFM-01936/S04/B0	8	1.733e-14	1.662e-14	5.107e-09	0
ZFM-01936/S04/B1	304	3.902e-06	1.427e-16	1.293e-08	260
ZFM-01936/S04/B2	336	4.21e-06	1.563e-16	1.294e-08	267
ZFM-01936/S04/B3	528	1.482e-06	2.776e-16	1.279e-08	38

Mouse/case/configuration	Free q	PG	AD error	FD max	Iter
ZFM-01936/S04/B4	560	6.785e-06	1.266e-16	1.292e-08	99
ZFM-01936/S05/B0	8	6.741e-15	6.529e-15	2.558e-09	0
ZFM-01936/S05/B1	304	9.098e-07	8.197e-17	7.282e-09	190
ZFM-01936/S05/B2	336	1.655e-06	1.206e-16	7.273e-09	191
ZFM-01936/S05/B3	528	1.915e-06	5.551e-16	6.664e-09	34
ZFM-01936/S05/B4	560	7.323e-07	1.006e-16	6.992e-09	71
ZFM-01936/S06/B0	8	4.363e-15	4.777e-15	2.567e-09	0
ZFM-01936/S06/B1	304	1.345e-06	1.002e-16	7.28e-09	192
ZFM-01936/S06/B2	336	9.31e-07	9.584e-17	7.284e-09	194
ZFM-01936/S06/B3	528	1.599e-06	1.874e-16	6.665e-09	37
ZFM-01936/S06/B4	560	7.346e-07	1.37e-16	7e-09	71
ZFM-01936/S07/B0	8	5.736e-16	5.934e-16	6.661e-10	0
ZFM-01936/S07/B1	304	6.873e-07	2.407e-17	1.378e-09	64
ZFM-01936/S07/B2	336	9.446e-07	3.112e-17	1.376e-09	62
ZFM-01936/S07/B3	528	4.738e-07	8.327e-17	9.408e-10	24
ZFM-01936/S07/B4	560	9.686e-07	3.123e-17	1.243e-09	27
ZFM-01936/S08/B0	8	4.363e-15	4.777e-15	2.567e-09	0
ZFM-01936/S08/B1	304	1.345e-06	1.002e-16	7.28e-09	192
ZFM-01936/S08/B2	336	9.31e-07	9.584e-17	7.284e-09	194
ZFM-01936/S08/B3	528	1.599e-06	1.874e-16	6.665e-09	37
ZFM-01936/S08/B4	560	7.346e-07	1.37e-16	7e-09	71
ZFM-01936/S09/B0	8	9.387e-15	8.776e-15	2.558e-09	0
ZFM-01936/S09/B1	304	4.327e-06	1.228e-16	7.269e-09	187
ZFM-01936/S09/B2	336	1.139e-06	9.458e-17	7.257e-09	186
ZFM-01936/S09/B3	528	3.949e-06	3.261e-16	6.651e-09	35
ZFM-01936/S09/B4	560	9.272e-07	1.18e-16	6.969e-09	70
ZFM-01936/S10/B0	8	4.363e-15	4.777e-15	2.567e-09	0
ZFM-01936/S10/B1	336	1.537e-06	6.939e-17	7.298e-09	200
ZFM-01936/S10/B2	368	1.212e-06	1.128e-16	7.284e-09	202
ZFM-01936/S10/B3	560	1.068e-06	2.845e-16	6.664e-09	39
ZFM-01936/S10/B4	592	3.648e-06	9.714e-17	6.995e-09	73
ZFM-01936/S11/B0	5	4.363e-15	4.777e-15	2.554e-09	0
ZFM-01936/S11/B1	190	8.909e-07	8.544e-17	6.783e-09	182
ZFM-01936/S11/B2	210	9.648e-07	7.644e-17	6.785e-09	177
ZFM-01936/S11/B3	270	8.103e-07	3.088e-16	6.621e-09	29
ZFM-01936/S11/B4	290	8.901e-07	1.301e-16	6.758e-09	67

S9. All contrasts, animal-level effects and uncertainty

Mouse is the independent statistical unit; units/trials/probes are not independent replicates. Gains retain full LL, exposure and actual scored nominal unit-bin denominators. Fixed contrast order is primary B4-B1, own B4-B3, other B4-B2, D-B4, C-B4, D-S. Development and Final are never pooled. The following two tables give original exact saved values; allFinal rows remain DESCRIPTIVE ONLY. Original CSV order is preserved separately in source copies, while this reporting view follows the original hypothesis order.

Development: exploratory; 10 mice from eight laboratories.

Contrast	Mean gain	95% whole-mouse CI	Raw p	Holm5 p
B4-B1	0.1125291851306621	[0.06745810552609997, 0.16368155129436163]	0.0009765625	not in family
B4-B3	0.03930120524139896	[0.01734158461894405, 0.06278710159999801]	0.01171875	0.05859375
B4-B2	0.026306579812766885	[0.0050302065057874785, 0.05501082886164163]	0.044921875	0.1796875
D-B4	-0.01477973215700219	[-0.03601613299004298, 0.0021497034444591255]	0.19921875	0.59765625
C-B4	0.00010824211289161302	[-0.000400099229921444, 0.000568692298035778]	0.685546875	0.859375
D-S	-0.0019684110379102277	[-0.005914920527199967 4, 0.00047306207126944814]	0.4296875	0.859375

Final: DESCRIPTIVE ONLY; six mice from four laboratories.

Contrast	Mean gain	95% whole-mouse CI	Raw p	Holm5 p
B4-B1	0.08658239652342035	[0.04944033234175033, 0.14110164811873427]	0.015625	not in family
B4-B3	0.05208972400526927	[0.03441612054329819, 0.07346442266178636]	0.03125	0.15625
B4-B2	0.006353386392021677	[-0.008903428833930957, 0.022405637931404047]	0.53125	1.0
D-B4	0.00235862122254638	[-0.003948272203674794, 0.009196280435455508]	0.53125	1.0
C-B4	0.00013410738971923112	[-5.783408566630739e-05, 0.0003429179830851248 5]	0.28125	1.0
D-S	0.0016697806796088692	[-0.00030445108225563776, 0.0051111034674382804]	0.5625	1.0

Original tests enumerate all2^n sign vectors. Primary uses mean(s*d)>=mean(d); secondary uses absolute means, with the frozen1e-14equality stabilization. No Monte Carlo correction is added to the exhaustive test. Interpretation requires independent sign-exchangeable symmetric zero-centered effects, not an arbitrary unrestricted null mean. There are64signs at n6, so minimum one-/two-sided p=0.015625/0.03125, and smallest Holm5secondary p=0.15625. Missing families retain p1slots. Holm adjustment is the cumulative maximum of ordered (5-rank+1)*p, capped1, with stable original-slot restoration.

The10,000whole-mouse draws use PCG64seed150101, arithmetic means, linear0.025/0.975quantiles. Each contrast reuses its specified fixed index stream. SD has n-1denominator; IQR uses linear quartiles. Original leave-one-mouse/lab, labmeans andequal-lab mean are saved diagnostics. The lab-block bootstrap uses seed150102, draws the original number of labs with replacement, and aggregates included mice's effect sums divided by included mouse counts; it is not an equal-lab primary estimator. Its small lab denominator is disclosed. No new cross-cohort or own-versus-other test is added by this supplement.

Cohort	Mouse	Contrast	Effect	Scored unit-bins
Dev	KS046	B4-B1	0.08052120076	9520
Dev	PL033	B4-B1	0.01895552682	9120
Dev	SWC_043	B4-B1	0.0297892462	7920
Dev	SWC_038	B4-B1	0.06548543728	12800
Dev	CSH_ZAD_026	B4-B1	0.2304086549	12960
Dev	NYU-30	B4-B1	0.0600708051	10000
Dev	KS051	B4-B1	0.264325883	6080
Dev	ZFM-01937	B4-B1	0.1345940559	6960
Dev	CSH_ZAD_019	B4-B1	0.1551709411	9200
Dev	UCLA035	B4-B1	0.08597010028	13760
Dev	KS046	B4-B3	0.05819635041	9520
Dev	PL033	B4-B3	0.01235773129	9120
Dev	SWC_043	B4-B3	0.02035371395	7920
Dev	SWC_038	B4-B3	0.03707496984	12800
Dev	CSH_ZAD_026	B4-B3	-0.02603455397	12960
Dev	NYU-30	B4-B3	0.05204759639	10000
Dev	KS051	B4-B3	0.01132994605	6080
Dev	ZFM-01937	B4-B3	0.03571423848	6960
Dev	CSH_ZAD_019	B4-B3	0.07860768589	9200
Dev	UCLA035	B4-B3	0.1133643741	13760
Dev	KS046	B4-B2	0.01828854523	9520
Dev	PL033	B4-B2	-0.02051362974	9120
Dev	SWC_043	B4-B2	0.007251290257	7920
Dev	SWC_038	B4-B2	0.01009090225	12800
Dev	CSH_ZAD_026	B4-B2	0.1380739421	12960
Dev	NYU-30	B4-B2	0.01063712653	10000
Dev	KS051	B4-B2	0.04754134899	6080
Dev	ZFM-01937	B4-B2	0.04220873531	6960
Dev	CSH_ZAD_019	B4-B2	0.01100047266	9200
Dev	UCLA035	B4-B2	-0.00151293544	13760
Dev	KS046	D-B4	0.004160042777	9520
Dev	PL033	D-B4	-0.03870847584	9120

Cohort	Mouse	Contrast	Effect	Scored unit-bins
Dev	SWC_043	D-B4	-0.009813369484	7920
Dev	SWC_038	D-B4	0.002351203746	12800
Dev	CSH_ZAD_026	D-B4	0.01855069352	12960
Dev	NYU-30	D-B4	-0.0011866799	10000
Dev	KS051	D-B4	-0.09508395869	6080
Dev	ZFM-01937	D-B4	-0.03552402696	6960
Dev	CSH_ZAD_019	D-B4	0.008564880997	9200
Dev	UCLA035	D-B4	-0.00110763174	13760
Dev	KS046	C-B4	-0.000195086033	9520
Dev	PL033	C-B4	0.001048917521	9120
Dev	SWC_043	C-B4	-4.082760353e-05	7920
Dev	SWC_038	C-B4	-0.0002416324679	12800
Dev	CSH_ZAD_026	C-B4	0.0008202170878	12960
Dev	NYU-30	C-B4	0.0001353345891	10000
Dev	KS051	C-B4	0.0001550140577	6080
Dev	ZFM-01937	C-B4	-0.001637741622	6960
Dev	CSH_ZAD_019	C-B4	0.001298190818	9200
Dev	UCLA035	C-B4	-0.0002599652185	13760
Dev	KS046	D-S	0.0002515342479	9520
Dev	PL033	D-S	0.0007118957586	9120
Dev	SWC_043	D-S	8.254664174e-08	7920
Dev	SWC_038	D-S	-0.001757812083	12800
Dev	CSH_ZAD_026	D-S	-0.01867244828	12960
Dev	NYU-30	D-S	0.0001041095974	10000
Dev	KS051	D-S	-0.001937933648	6080
Dev	ZFM-01937	D-S	0.002402706443	6960
Dev	CSH_ZAD_019	D-S	-0.0005221152163	9200
Dev	UCLA035	D-S	-0.0002641297404	13760
Fin	UCLA037	B4-B1	0.09113110645	13680
Fin	DY_014	B4-B1	0.0747880607	17440
Fin	ZFM-01576	B4-B1	0.215114199	32160
Fin	NR_0027	B4-B1	0.03500649848	15280
Fin	DY_011	B4-B1	0.05506809263	12320
Fin	ZFM-01936	B4-B1	0.04838642187	17920
Fin	UCLA037	B4-B3	0.04017741117	13680
Fin	DY_014	B4-B3	0.06586010317	17440
Fin	ZFM-01576	B4-B3	0.04838456595	32160
Fin	NR_0027	B4-B3	0.02591083568	15280
Fin	DY_011	B4-B3	0.1002103437	12320

Cohort	Mouse	Contrast	Effect	Scored unit-bins
Fin	ZFM-01936	B4-B3	0.03199508434	17920
Fin	UCLA037	B4-B2	0.009146492357	13680
Fin	DY_014	B4-B2	0.002673076108	17440
Fin	ZFM-01576	B4-B2	0.0390199347	32160
Fin	NR_0027	B4-B2	0.008926726412	15280
Fin	DY_011	B4-B2	-0.02472228929	12320
Fin	ZFM-01936	B4-B2	0.003076378064	17920
Fin	UCLA037	D-B4	0.008419415286	13680
Fin	DY_014	D-B4	0.01398749592	17440
Fin	ZFM-01576	D-B4	0.01000539177	32160
Fin	NR_0027	D-B4	-0.005457865318	15280
Fin	DY_011	D-B4	-0.007593098119	12320
Fin	ZFM-01936	D-B4	-0.005209612205	17920
Fin	UCLA037	C-B4	9.305915588e-05	13680
Fin	DY_014	C-B4	8.682844193e-05	17440
Fin	ZFM-01576	C-B4	0.0005426344254	32160
Fin	NR_0027	C-B4	-0.0002606538652	15280
Fin	DY_011	C-B4	0.0003156948268	12320
Fin	ZFM-01936	C-B4	2.708135353e-05	17920
Fin	UCLA037	D-S	0.01005989983	13680
Fin	DY_014	D-S	-0.0005442196321	17440
Fin	ZFM-01576	D-S	0.001031289859	32160
Fin	NR_0027	D-S	-0.0001591523655	15280
Fin	DY_011	D-S	-0.0003689846995	12320
Fin	ZFM-01936	D-S	-1.489151692e-07	17920

S10. All eleven baseline-only sensitivity panels

Only B0-B4 were sensitivity-refitted. D/C/S have primary-configuration-only evidence. All source/QC/ranking transforms were A/C-defined before later scoring. S01/S04-S11 are nine mandatory cases; S02/S03 are capacity-conditional panels, retained even when unavailable. The frozen rules are:

Case	Frozen intervention	Interpretation and coverage
S01	First5 qualified ranked units	Mandatory minimum6pairedFinal mice
S02	First12 of same ranking	Capacity-conditional minimum6
S03	First20 of same ranking	Final4mice; aggregateunavailable
S04	100-ms bins; physical historywindows;5score bins	Different nats/bin scale; not directly comparable as an equal rate
S05	History edges0,0.05,0.1,0.2,0.25s;0.25s neural warm-up	Narrower history estimand; behavior supportunchanged

Case	Frozen intervention	Interpretation and coverage
S06	Rate1-50Hz; ISI<=0.0025;>=100spikes; quartile>=0.2Hz	First8 under original training-only ranking
S07	p=min(1,2/r) per spike from unthinnedA/C rate	Independent role-local stream; expected2Hzcap, not exact matching
S08	Known saturation endpoints dilated±10ms	No unknown-artifact certification
S09	Known saturation endpoints dilated±50ms	No source-missing interval invented
S10	Trial/session cubic+quartic drift terms	41 nuisance columns; not added behavioral sensors
S11	First5qualified units with target±2nativecontact indicesallMOp	Anatomical boundaryproxy; not histologicaldistance

S07 uses the first eight bytes, big-endian, of SHA256 of UTF8 `P15-M1|150108|S07|subject|unit_uuid|role`, using PCG64. One uniform is consumed per source spike in original native within-role row order, including spikes outside retained epochs; duplicate timestamps retain source order. Retain iff uniform<p. A, B, E, C and D streams are independent and processed only when authorized. There is one realization, with no favorable redraw. Score data do not change eligibility, rank or retention probability. Achieved A/C rate evidence remains in original case states; D rates were not used for matching.

Development has 33 contrast-by-case summaries. Final has 33 summaries and 66 mouse-case states: 64 available and two capacity-unavailable. Final S02 has six mice; S03 has four. All nine mandatory Final cases have six paired mice and positive primary/own/other mean gains, without repairing the failed cohort, assumption or precision conditions. The two full tables below retain coverage, status, mean and interval, including unavailable entries.

Case	Contrast	Mice	Mean	95% whole-mouse CI	Status
S01	B4-B1	10	0.12195633	[0.067087561,0.18925188]	DESCRIPTIVE_SENSITIVITY
S01	B4-B3	10	0.055910131	[0.016218275,0.096893846]	DESCRIPTIVE_SENSITIVITY
S01	B4-B2	10	0.024848328	[-0.0026289962,0.059608829]	DESCRIPTIVE_SENSITIVITY
S02	B4-B1	9	0.11871086	[0.06663047,0.17904306]	DESCRIPTIVE_SENSITIVITY
S02	B4-B3	9	0.037484085	[0.02667183,0.050808617]	DESCRIPTIVE_SENSITIVITY
S02	B4-B2	9	0.027270244	[0.00659337,0.057527593]	DESCRIPTIVE_SENSITIVITY
S03	B4-B1	6	0.13378186	[0.077903632,0.1986272]	DESCRIPTIVE_SENSITIVITY
S03	B4-B3	6	0.03422748	[0.020131188,0.052595157]	DESCRIPTIVE_SENSITIVITY
S03	B4-B2	6	0.038781023	[0.012669258,0.085918538]	DESCRIPTIVE_SENSITIVITY
S04	B4-B1	10	0.23351777	[0.12665491,0.35957399]	DESCRIPTIVE_SENSITIVITY
S04	B4-B3	10	0.04918818	[0.0032783582,0.096177838]	DESCRIPTIVE_SENSITIVITY
S04	B4-B2	10	0.057045412	[0.01142023,0.11414502]	DESCRIPTIVE_SENSITIVITY
S05	B4-B1	10	0.098553662	[0.062695405,0.13913796]	DESCRIPTIVE_SENSITIVITY
S05	B4-B3	10	0.035406593	[0.021129919,0.04922488]	DESCRIPTIVE_SENSITIVITY
S05	B4-B2	10	0.026534894	[0.0099243809,0.048079332]	DESCRIPTIVE_SENSITIVITY
S06	B4-B1	10	0.11258296	[0.067513227,0.16372615]	DESCRIPTIVE_SENSITIVITY
S06	B4-B3	10	0.039130592	[0.017141294,0.0626932]	DESCRIPTIVE_SENSITIVITY

Case	Contrast	Mice	Mean	95% whole-mouse CI	Status
S06	B4-B2	10	0.026367954	[0.0050640205,0.055061618]	DESCRIPTIVE_SENSITIVITY
S07	B4-B1	10	0.025252439	[0.01258791,0.040195612]	DESCRIPTIVE_SENSITIVITY
S07	B4-B3	10	0.010126573	[0.0023467699,0.017613963]	DESCRIPTIVE_SENSITIVITY
S07	B4-B2	10	0.0089023893	[0.0022852183,0.016294652]	DESCRIPTIVE_SENSITIVITY
S08	B4-B1	10	0.11251173	[0.06718657,0.16381417]	DESCRIPTIVE_SENSITIVITY
S08	B4-B3	10	0.039291897	[0.017355773,0.062769816]	DESCRIPTIVE_SENSITIVITY
S08	B4-B2	10	0.026221412	[0.0049920568,0.054839626]	DESCRIPTIVE_SENSITIVITY
S09	B4-B1	10	0.11301509	[0.06718638,0.16462682]	DESCRIPTIVE_SENSITIVITY
S09	B4-B3	10	0.039287748	[0.017363144,0.062733514]	DESCRIPTIVE_SENSITIVITY
S09	B4-B2	10	0.026798957	[0.0049954436,0.056555106]	DESCRIPTIVE_SENSITIVITY
S10	B4-B1	10	0.11066839	[0.061743828,0.16774678]	DESCRIPTIVE_SENSITIVITY
S10	B4-B3	10	0.037996857	[0.019582701,0.057262161]	DESCRIPTIVE_SENSITIVITY
S10	B4-B2	10	0.028352243	[0.0044851042,0.058368412]	DESCRIPTIVE_SENSITIVITY
S11	B4-B1	10	0.12195633	[0.067087561,0.18925188]	DESCRIPTIVE_SENSITIVITY
S11	B4-B3	10	0.055910131	[0.016218275,0.096893846]	DESCRIPTIVE_SENSITIVITY
S11	B4-B2	10	0.024848328	[-0.0026289962,0.059608829]	DESCRIPTIVE_SENSITIVITY

Case	Contrast	Mice	Mean	95% whole-mouse CI	Status
S01	B4-B1	6	0.093219623	[0.059477353,0.13380611]	DESCRIPTIVE_SENSITIVITY
S01	B4-B3	6	0.076281462	[0.044802215,0.11470378]	DESCRIPTIVE_SENSITIVITY
S01	B4-B2	6	0.0045942683	[-0.0015574551,0.01186073]	DESCRIPTIVE_SENSITIVITY
S02	B4-B1	6	0.06849713	[0.041465025,0.10483097]	DESCRIPTIVE_SENSITIVITY
S02	B4-B3	6	0.041036095	[0.028188494,0.055409972]	DESCRIPTIVE_SENSITIVITY
S02	B4-B2	6	0.0040374913	[-0.011634553,0.015923449]	DESCRIPTIVE_SENSITIVITY
S03	B4-B1	4	0.078832882	[0.060301449,0.097364315]	UNAVAILABLE_MINIMUM_PAIRED_MICE
S03	B4-B3	4	0.033746473	[0.026167564,0.041325382]	UNAVAILABLE_MINIMUM_PAIRED_MICE
S03	B4-B2	4	0.0063502718	[0.00073700623,0.011963537]	UNAVAILABLE_MINIMUM_PAIRED_MICE
S04	B4-B1	6	0.16734705	[0.090073244,0.29239782]	DESCRIPTIVE_SENSITIVITY
S04	B4-B3	6	0.083931258	[0.053808511,0.12364046]	DESCRIPTIVE_SENSITIVITY
S04	B4-B2	6	0.0098894161	[-0.011455824,0.036671838]	DESCRIPTIVE_SENSITIVITY
S05	B4-B1	6	0.11767008	[0.046498266,0.23460058]	DESCRIPTIVE_SENSITIVITY
S05	B4-B3	6	0.036887241	[0.0043585151,0.066972102]	DESCRIPTIVE_SENSITIVITY
S05	B4-B2	6	0.03219263	[-0.010758796,0.10040166]	DESCRIPTIVE_SENSITIVITY
S06	B4-B1	6	0.085112878	[0.049440332,0.14055086]	DESCRIPTIVE_SENSITIVITY
S06	B4-B3	6	0.053427779	[0.035430162,0.074397594]	DESCRIPTIVE_SENSITIVITY
S06	B4-B2	6	0.0062747119	[-0.0089821034,0.022405638]	DESCRIPTIVE_SENSITIVITY
S07	B4-B1	6	0.013317139	[0.0027729692,0.027086531]	DESCRIPTIVE_SENSITIVITY
S07	B4-B3	6	0.0078230962	[-0.002891691,0.020533788]	DESCRIPTIVE_SENSITIVITY

Case	Contrast	Mice	Mean	95% whole-mouse CI	Status
S07	B4-B2	6	0.0039684782	[-0.0014030408,0.010879155]	DESCRIPTIVE_SENSITIVITY
S08	B4-B1	6	0.086537756	[0.049478607,0.14103787]	DESCRIPTIVE_SENSITIVITY
S08	B4-B3	6	0.052026846	[0.034383137,0.073453428]	DESCRIPTIVE_SENSITIVITY
S08	B4-B2	6	0.006254927	[-0.0090018882,0.022405638]	DESCRIPTIVE_SENSITIVITY
S09	B4-B1	6	0.08701024	[0.050024445,0.14144132]	DESCRIPTIVE_SENSITIVITY
S09	B4-B3	6	0.051676631	[0.034020273,0.073359882]	DESCRIPTIVE_SENSITIVITY
S09	B4-B2	6	0.0066430713	[-0.008623008,0.022414902]	DESCRIPTIVE_SENSITIVITY
S10	B4-B1	6	0.085416426	[0.049748955,0.13890595]	DESCRIPTIVE_SENSITIVITY
S10	B4-B3	6	0.048770873	[0.032664198,0.066353752]	DESCRIPTIVE_SENSITIVITY
S10	B4-B2	6	0.006568526	[-0.0070979301,0.021792904]	DESCRIPTIVE_SENSITIVITY
S11	B4-B1	6	0.093219623	[0.059477353,0.13380611]	DESCRIPTIVE_SENSITIVITY
S11	B4-B3	6	0.076281462	[0.044802215,0.11470378]	DESCRIPTIVE_SENSITIVITY
S11	B4-B2	6	0.0045942683	[-0.0015574551,0.01186073]	DESCRIPTIVE_SENSITIVITY

S11. Four 999-replicate finite-reference suites and coverage

Each available suite uses 999 frozen reference realizations per mouse and score role, with no null model refitting or recursive count generation. Targets, unit identities, trial assignments, task covariates, valid masks and fitted states remain fixed. A reference alters specified history or wheel inputs. All contrasts reuse the same realization in a bundle.

N1 rotates each 20-bin multivariate warm-up-plus-score sequence by a nonzero offset from 1 to 19, with all units together. Epoch/unit totals are preserved while neural/task alignment changes. History and C features are recomputed; D updates its filter from surrogate past counts while predicting original targets. N2 cyclically shifts history donors within stimulus-side, contrast, order-half and wheel strata, without self-donors. N3 substitutes wheel columns 24-37, omits wheel from the strata, and reports B4 likelihood reliance plus descriptive B4-B1 changes. It is not a separate D/C wheel null. N4 shifts complete five-trial history blocks, preserving within-block order and population synchrony.

N2-N4 require at least 60% eligible-epoch coverage and 30 control epochs per mouse; N4 needs at least two complete blocks. Singleton strata and incomplete blocks are excluded from both original and control exposures, not from primary scoring. No post-score coarsening is allowed. N2's wheel threshold is the ordinary A/C median of absolute strictly past 0.5-s pre-epoch displacement, with ties assigned lower. Side, contrast threshold 0.25 and original role half are metadata-defined. Continuous nuisance imbalance can remain; N4 is not an exact drift-independent counterfactual.

The RNG is PCG64. Its seed is the first eight bytes, big-endian, of SHA256 of UTF8 `P15-M1|150107|suite|subject|partition|replicate`. N1 draws in original trial order; N2/N3 visit lexicographically ordered strata and sorted original trial ordinals; N4 visits complete blocks in original block order. No offset is drawn for a skipped singleton or incomplete panel. No wall-clock or Python-hash seed, replacement seed or repetition is permitted.

For an available suite, p_ref=(1+number of reference means at least the observed mean)/1000 on the same complete control-mouse cohort and exposures. Holm4 retains all four slots separately for each contrast; an unavailable suite has p=1. These are finite destructive references, not exact population, conditional or causal null p-values. Descriptive gates jointly require at least six paired mice, at least ceil(0.75n) above their own reference median, and positive equal-mouse mean degradation. A small reference p does not clear a failed component. N3's repeated B4-likelihood diagnostic is explicitly labelled and is not an additional independent test for each model.

Final N1/N2/N3 each covered six mice. N4 covered only ZFM-01576, DY_011 and ZFM-01936, so its aggregate was unavailable. The Final batch contained 21*999=20,979 available bundles and 24 suite journals, including three unavailable mouse-suite panels. Development N4 covered only five mice: KS046, SWC_043, SWC_038, KS051 and ZFM-01937. The following 48 rows retain six contrasts times four suites in both cohorts. Exact original p-values and statuses remain in the copied CSVs; the coverage table lists each mouse's original availability and coverage.

Cohort	Contrast	Suite	Mice	Above median	Descriptive gate	p_ref	Holm4
Dev	B4-B1	N1	10	7	False	0.001	0.004
Dev	B4-B1	N2	10	10	True	0.001	0.004
Dev	B4-B1	N3	10	10	wheel LL diagnostic	0.001	0.004
Dev	B4-B1	N4	5	unavailable	unavailable	1.0	1.0
Dev	B4-B3	N1	10	7	False	0.001	0.004
Dev	B4-B3	N2	10	10	True	0.001	0.004
Dev	B4-B3	N3	10	10	wheel LL diagnostic	0.001	0.004
Dev	B4-B3	N4	5	unavailable	unavailable	1.0	1.0
Dev	B4-B2	N1	10	9	True	0.001	0.004
Dev	B4-B2	N2	10	9	True	0.001	0.004
Dev	B4-B2	N3	10	10	wheel LL diagnostic	0.001	0.004
Dev	B4-B2	N4	5	unavailable	unavailable	1.0	1.0
Dev	D-B4	N1	10	3	False	0.901	1.0
Dev	D-B4	N2	10	7	False	0.001	0.004
Dev	D-B4	N3	10	10	wheel LL diagnostic	0.001	0.004
Dev	D-B4	N4	5	unavailable	unavailable	1.0	1.0
Dev	C-B4	N1	10	7	False	0.001	0.004
Dev	C-B4	N2	10	10	True	0.001	0.004
Dev	C-B4	N3	10	10	wheel LL diagnostic	0.001	0.004
Dev	C-B4	N4	5	unavailable	unavailable	1.0	1.0
Dev	D-S	N1	10	7	False	0.001	0.004
Dev	D-S	N2	10	5	False	0.059	0.118

Cohort	Contrast	Suite	Mice	Above median	Descriptive gate	p_ref	Holm4
Dev	D-S	N3	10	10	wheel LL diagnostic	0.001	0.004
Dev	D-S	N4	5	unavailable	unavailable	1.0	1.0
Fin	B4-B1	N1	6	6	True	0.001	0.004
Fin	B4-B1	N2	6	6	True	0.001	0.004
Fin	B4-B1	N3	6	6	wheel LL diagnostic	0.001	0.004
Fin	B4-B1	N4	3	unavailable	unavailable	1.0	1.0
Fin	B4-B3	N1	6	6	True	0.001	0.004
Fin	B4-B3	N2	6	6	True	0.001	0.004
Fin	B4-B3	N3	6	6	wheel LL diagnostic	0.001	0.004
Fin	B4-B3	N4	3	unavailable	unavailable	1.0	1.0
Fin	B4-B2	N1	6	5	True	0.001	0.004
Fin	B4-B2	N2	6	5	True	0.001	0.004
Fin	B4-B2	N3	6	6	wheel LL diagnostic	0.001	0.004
Fin	B4-B2	N4	3	unavailable	unavailable	1.0	1.0
Fin	D-B4	N1	6	5	True	0.001	0.004
Fin	D-B4	N2	6	4	False	0.871	1.0
Fin	D-B4	N3	6	6	wheel LL diagnostic	0.001	0.004
Fin	D-B4	N4	3	unavailable	unavailable	1.0	1.0
Fin	C-B4	N1	6	5	True	0.001	0.004
Fin	C-B4	N2	6	5	True	0.001	0.004
Fin	C-B4	N3	6	6	wheel LL diagnostic	0.001	0.004
Fin	C-B4	N4	3	unavailable	unavailable	1.0	1.0
Fin	D-S	N1	6	4	False	0.001	0.004
Fin	D-S	N2	6	2	False	0.622	1.0
Fin	D-S	N3	6	6	wheel LL diagnostic	0.001	0.004
Fin	D-S	N4	3	unavailable	unavailable	1.0	1.0

Cohort	Mouse	Suite	Eligible / total epochs	Available
Dev	CSH_ZAD_019	N1	115 / 115	True
Dev	CSH_ZAD_019	N2	112 / 115	True
Dev	CSH_ZAD_019	N3	115 / 115	True
Dev	CSH_ZAD_019	N4	65 / 115	False
Dev	CSH_ZAD_026	N1	162 / 162	True
Dev	CSH_ZAD_026	N2	160 / 162	True

Cohort	Mouse	Suite	Eligible / total epochs	Available
Dev	CSH_ZAD_026	N3	162 / 162	True
Dev	CSH_ZAD_026	N4	50 / 162	False
Dev	KS046	N1	119 / 119	True
Dev	KS046	N2	119 / 119	True
Dev	KS046	N3	119 / 119	True
Dev	KS046	N4	75 / 119	True
Dev	KS051	N1	76 / 76	True
Dev	KS051	N2	74 / 76	True
Dev	KS051	N3	75 / 76	True
Dev	KS051	N4	65 / 76	True
Dev	NYU-30	N1	125 / 125	True
Dev	NYU-30	N2	123 / 125	True
Dev	NYU-30	N3	125 / 125	True
Dev	NYU-30	N4	50 / 125	False
Dev	PL033	N1	114 / 114	True
Dev	PL033	N2	111 / 114	True
Dev	PL033	N3	114 / 114	True
Dev	PL033	N4	65 / 114	False
Dev	SWC_038	N1	160 / 160	True
Dev	SWC_038	N2	159 / 160	True
Dev	SWC_038	N3	160 / 160	True
Dev	SWC_038	N4	115 / 160	True
Dev	SWC_043	N1	99 / 99	True
Dev	SWC_043	N2	98 / 99	True
Dev	SWC_043	N3	99 / 99	True
Dev	SWC_043	N4	70 / 99	True
Dev	UCLA035	N1	172 / 172	True
Dev	UCLA035	N2	171 / 172	True
Dev	UCLA035	N3	172 / 172	True
Dev	UCLA035	N4	90 / 172	False
Dev	ZFM-01937	N1	87 / 87	True
Dev	ZFM-01937	N2	83 / 87	True
Dev	ZFM-01937	N3	86 / 87	True
Dev	ZFM-01937	N4	80 / 87	True
Fin	DY_011	N1	154 / 154	True
Fin	DY_011	N2	151 / 154	True
Fin	DY_011	N3	154 / 154	True
Fin	DY_011	N4	100 / 154	True

Cohort	Mouse	Suite	Eligible / total epochs	Available
Fin	DY_014	N1	218 / 218	True
Fin	DY_014	N2	218 / 218	True
Fin	DY_014	N3	218 / 218	True
Fin	DY_014	N4	105 / 218	False
Fin	NR_0027	N1	191 / 191	True
Fin	NR_0027	N2	189 / 191	True
Fin	NR_0027	N3	191 / 191	True
Fin	NR_0027	N4	30 / 191	False
Fin	UCLA037	N1	171 / 171	True
Fin	UCLA037	N2	168 / 171	True
Fin	UCLA037	N3	171 / 171	True
Fin	UCLA037	N4	60 / 171	False
Fin	ZFM-01576	N1	402 / 402	True
Fin	ZFM-01576	N2	402 / 402	True
Fin	ZFM-01576	N3	402 / 402	True
Fin	ZFM-01576	N4	395 / 402	True
Fin	ZFM-01936	N1	224 / 224	True
Fin	ZFM-01936	N2	224 / 224	True
Fin	ZFM-01936	N3	224 / 224	True
Fin	ZFM-01936	N4	215 / 224	True

S12. Failed conditions and exact scientific verdicts

The primary positive-claim conjunction required at least eight Final mice and four labs, no lab over 50%, defensible independence/sign symmetry, valid source/numerical gates, mean at least 0.001, positive CI lower bound, width at most 0.01, one-sided p at most 0.05, at least 75% positive mice, positive leave-one-mouse/lab-out means, all nine mandatory sensitivities with six mice and positive means, and N1/N2 descriptive gates. Final failed its cohort, width and confirmation-assumption requirements despite positive effects and passing primary N1/N2 descriptive conditions. Its mandatory label remains DESCRIPTIVE ONLY.

Secondary conditions add Holm5 two-sided p at most 0.05 and the same cohort, precision and stability requirements. Own/other history require their nine baseline sensitivities and N1/N2; D-B4 and D-S add N4; C-B4 requires N1/N2 without borrowing baseline sensitivity credit; D-S also requires its paired S state. Persistent-state wording would require both D-B4 and D-S conjunctions and the IID identity/N4 conditions. None was met. All five Holm5 tests were nonsignificant. No equivalence, absence of biology, model winner or biological-state claim follows.

Development primary N1 had 7/10, below the required eight, and CI width 0.0962234458 exceeded 0.01; N2 had 10/10. Final primary N1/N2 at 6/6 do not erase that failure. Final D-B4 N2 had 4/6 above median and negative mean degradation. Final D-S N1 had 4/6,

below five; N2 had 2/6 and negative degradation. N4 aggregate was unavailable in both cohorts, not negative or passed. These failures also appear in the abstract, Results, Discussion and Limitations.

S13. Animal/laboratory heterogeneity and anatomy

All 16 admitted mice's effects are listed above. The following table gives all six contrasts in each cohort, with minimum, maximum, median, SD, positive count, equal-lab mean and lab-block CI. The companion CSV/JSON retains original leave-out vectors and lab means with their original order. These are previously registered saved diagnostics, not new hypothesis tests. Main Figure 7 describes native anatomy and Final lab means without an added association p-value.

Cohort	Contrast	Min / max	Median	SD	Positive	Equal-lab mean	Lab-block CI95
Dev	B4-B1	0.0189555 / 0.264326	0.0832457	0.0826907	10	0.0950098	[0.0545127,0.154958]
Dev	B4-B3	-0.0260346 / 0.113364	0.0363946	0.0390211	9	0.0414953	[0.0259242,0.0605343]
Dev	B4-B2	-0.0205136 / 0.138074	0.0108188	0.0438836	8	0.0194517	[0.00238277,0.0476615]
Dev	D-B4	-0.095084 / 0.0185507	-0.00114716	0.0335925	4	-0.0144866	[-0.0321446,0.00200076]
Dev	C-B4	-0.00163774 / 0.00129819	4.72535e-05	0.000832089	5	5.40665e-06	[-0.000486935,0.000612221]
Dev	D-S	-0.0186724 / 0.00240271	-0.000132024	0.00599496	5	-0.00115545	[-0.005144,0.000595433]
Fin	B4-B1	0.0350065 / 0.215114	0.0649281	0.0660188	6	0.080704	[0.0490377,0.118211]
Fin	B4-B3	0.0259108 / 0.10021	0.044281	0.027386	6	0.0473283	[0.0316224,0.0748746]
Fin	B4-B2	-0.0247223 / 0.0390199	0.00600155	0.0203544	5	0.00702419	[-0.00814302,0.0193165]
Fin	D-B4	-0.0075931 / 0.0139875	0.0016049	0.00946414	3	0.00213916	[-0.00199584,0.00601081]
Fin	C-B4	-0.000260654 / 0.000542634	8.99438e-05	0.000272426	5	7.96312e-05	[-8.37974e-05,0.000257458]
Fin	D-S	-0.00054422 / 0.0100599	-7.96506e-05	0.00414714	2	0.00248993	[-0.000357452,0.00624217]

Cohort	Mouse	Population	Units	Superficial	Middle	Deep	Root/unknown
Dev	KS046	PRIMARY8	8	0	0	8	0
Dev	KS046	S11_BOUNDARY5	5	0	0	5	0
Dev	PL033	PRIMARY8	8	0	0	8	0
Dev	PL033	S11_BOUNDARY5	5	0	0	5	0
Dev	SWC_043	PRIMARY8	8	0	0	8	0

Cohort	Mouse	Population	Units	Superficial	Middle	Deep	Root/unknown
Dev	SWC_043	S11_BOUNDARY5	5	0	0	5	0
Dev	SWC_038	PRIMARY8	8	0	0	8	0
Dev	SWC_038	S11_BOUNDARY5	5	0	0	5	0
Dev	CSH_ZAD_026	PRIMARY8	8	0	0	8	0
Dev	CSH_ZAD_026	S11_BOUNDARY5	5	0	0	5	0
Dev	NYU-30	PRIMARY8	8	0	0	8	0
Dev	NYU-30	S11_BOUNDARY5	5	0	0	5	0
Dev	KS051	PRIMARY8	8	4	0	4	0
Dev	KS051	S11_BOUNDARY5	5	3	0	2	0
Dev	ZFM-01937	PRIMARY8	8	3	0	5	0
Dev	ZFM-01937	S11_BOUNDARY5	5	3	0	2	0
Dev	CSH_ZAD_019	PRIMARY8	8	0	0	8	0
Dev	CSH_ZAD_019	S11_BOUNDARY5	5	0	0	5	0
Dev	UCLA035	PRIMARY8	8	5	0	3	0
Dev	UCLA035	S11_BOUNDARY5	5	3	0	2	0
Fin	UCLA037	PRIMARY8	8	2	0	6	0
Fin	UCLA037	S11_BOUNDARY5	5	1	0	4	0
Fin	DY_014	PRIMARY8	8	0	0	8	0
Fin	DY_014	S11_BOUNDARY5	5	0	0	5	0
Fin	ZFM-01576	PRIMARY8	8	1	0	7	0
Fin	ZFM-01576	S11_BOUNDARY5	5	1	0	4	0
Fin	NR_0027	PRIMARY8	8	0	0	8	0
Fin	NR_0027	S11_BOUNDARY5	5	0	0	5	0
Fin	DY_011	PRIMARY8	8	0	0	8	0
Fin	DY_011	S11_BOUNDARY5	5	0	0	5	0
Fin	ZFM-01936	PRIMARY8	8	1	0	7	0
Fin	ZFM-01936	S11_BOUNDARY5	5	0	0	5	0

S14. Access, leakage controls and one-shot integrity

Model policies, unit ranking, QC, scalers and sensitivity configurations used A/C only. Upstream whole-session provider labels are disclosed. M0/M1 statements about unaccessed Final data describe those historical stages; the later authorized M6 D assessment is now complete. Those old scope statements are not presented as current unexposed status.

Final C's trusted local reader had explicit authorization for underlying whole-record ALF decoding solely to classify/export C. It did not use D neural counts, finite/range summaries, scales, QC, selection or thinning streams for calibration. This is ordinary local access discipline, not third-party custody or protection against malicious administrators. Complete C states were sealed, and independent reviews bound the 546-input candidate before the one-shot gate.

The C seal SHA256 was `4a6fb543eaed33fb7ef58c01677eab520128de5f2b001a682c481d323b3e4c67`; pre-D candidate SHA256 was `d59df6d55cb904267ee4ed08e6dcfae187529cd6a170d6553f048860c0b743e2`; pre-D gate SHA256 was `1c66ae3519130b6cb94f890ddc485a0ab1a03f5ec9c9611af7d83fe79590cd82`. The completed D claim SHA256 was `95108b1b992ae69185f083826dc67b0a79765afec3ee195449afd49017a8b726`. Each is checked against the immutable archive, not generated from an assumed result.

The execution receipt and consumed claim record the complete six-mouse batch. Registered references belong to that one assessment, not alternative Final draws. Manuscript production cannot refit C/D models, select policies after D or repeat D. The original processing workflow is documented for audit; the paper's reproduction command cannot invoke it.

S15. Source-to-claim and A1-A10 coverage

`A1_A10_CLAIM_AUDIT.md` maps the frozen question, source/licence/cohort, recording/physiology, models, role isolation, effects/statistics, controls/sensitivities, one-shot integrity, negative findings and reproduction to saved artifacts and Main Figures 1-7/Supplement sections. `AUTHOR_SCIENTIFIC_REVIEW.md` checks all ten author questions A-J. These are internal consistency reviews, not external peer review. Authorship, funding, ethics and conflicts still require human author verification before external submission.

No causal connectivity, motor-command mechanism, all-movement independence, persistent biological states, discrete neural atoms, CWP/NAP, human/BCI validation or independent-laboratory replication is supported. A positive mean and small conditional p do not replace the failed registered conjunction.

S16. Reproduction, environment and source archive

Run from repository root: `powershell -NoProfile -ExecutionPolicy Bypass -File paper15_manuscript/reproduce.ps1`. On another Windows machine, install the exact document dependencies in `requirements.txt` and pass `-Python` with that interpreter. Poppler tools must be available for page rendering. `REPRODUCIBILITY.md` separates saved-result reproduction from independent raw-data fit reproduction.

The command verifies source hashes, denies network and NumPy array loading, blocks old scientific-package imports, checks original mouse-level means/sign flips/bootstrap/Holm arithmetic, recopies unchanged tables, renders seven figures and rebuilds both PDFs. It checks PDF text, page numbers, glyphs and private paths. It does not query neural datasets, fit a model, compute likelihoods or call a prediction worker. No new hypothesis or statistical family is introduced. Original processing and access documentation are an audit trail, not permission to reuse the consumed D assessment.

The scientific base is `bdec0e19621cc8018abe11c5633ce76aff611443`. All five scientific packages and source/review dependencies retain original hashes. Manuscript commits contain only the new package and its mission evidence/current daily prefix. Live current commit/remote/clean proof is external to its own tracked bytes, avoiding a recursive commit claim. Original failures remain immutable.

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